

Model Analysis

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1 Data analysis /Dataframe import and filters

1.1 DataFrame import and filters

The data has been imported with the following(s) function(s):

```
def build_df():
    if flag_IR:
        df, features_engineered = import_data_IR()
        if flag_IR_maskP:
            df = mask_period_IR(df)
        if flag_IR_balR:
            df = mask_balanceHC_random_IR(df, n_samples=None)
    if flag_ER:
        df, features_engineered = import_data_ER()

    cols_to_check = target_col + features_X
    mask = np.isfinite(df[cols_to_check]).all(axis=1)
    df = df[mask]
```

```

cols = list(dict.fromkeys(target_col + features_X +
                          feats_to_balance + feats_grouping +
                          feats_reference+ feats_postproc))

df = df[cols]
#df['campaign'] = "ALL"

return df

```

```

def import_data_IR():
    if target_col[0] == "i_NH3":
        file_path = "csv_decomposed/innova/df_signal_decomposition_NH3_junesept.csv"
        #file_path = "csv/innova/switched/df_signal_decomposition_10_50_200.csv"
    elif target_col[0] == "i_CO2":
        file_path = "csv_decomposed/innova/df_signal_decomposition_CO2_current.csv"
    else:
        raise ValueError("Target is not supported")

    df = pd.read_csv(file_path, sep=";")
    df["datetime"] = pd.to_datetime(df["datetime"])
    df.sort_values("datetime").reset_index(drop=True)

    if target_col[0] == "i_NH3":
        df["campaign"] = df["datetime"].apply(assign_campaign)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv_decomposed/innova/features_decomposition_NH3_list_junesept.json", "r") as f:
            #with open("csv/innova/switched/features_decomposition_list_10_50_200.json", "r") as f:
            dict_features = json.load(f)
    elif target_col[0] == "i_CO2":
        df["campaign"] = df["datetime"].apply(assign_campaign_iCO2)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point_iCO2, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv/innova/features_decomposition_CO2_list.json", "r") as f:
            dict_features = json.load(f)
    else:
        raise ValueError("The target col do not match any existing function")

    df["hot"] = df.apply(assign_hot_season,axis=1)
    df["umidity_range"] = df.apply(assign_humidity, axis = 1)

```

```
#Removing of rabbit 4
df["column"] = df.apply(assign_colonna, axis = 1)
df['height'] = df.apply(assign_altezza, axis = 1)

return df,dict_features
```

```
def assign_innova_point(row):
    camp = row["campaign"]
    rid = row["id_rabbit"]

    # if camp == "2025_Jun":
    #     mapping = {2: "sl11", 4: "sl5", 6: "sl4", 1: "sl3", 5: "no"}
    # elif camp == "2025_Sep":
    #     mapping = {2: "sl11", 4: "sl5", 3: "sl4", 6: "sl3", 5: "no"}
    if camp == "2025_Apr":
        #mapping = {1: "no", 2: "no", 4: "no", 6: "sl5"}
        #mapping = {1: "no", 2: "sl5", 4: "no", 6: "sl5"}
        mapping = {1: "no", 2: "no", 4: "no", 6: "no"}
    elif camp == "2025_Jun":
        mapping = {2: "sl11", 4: "sl5", 1: "sl3", 6: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 1: "sl4", 6: "sl3", 5: "no"}
    elif camp == "2025_Sep":
        mapping = {2: "sl11", 4: "sl5", 6: "sl3", 3: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 6: "sl4", 3: "sl3", 5: "no"}
    elif camp == "2026_Gen":
        #mapping = {1: "no", 4: "no", 5: "sl11", 6: "no"}
        #mapping = {1: "sl11", 4: "no", 5: "sl11", 6: "sl11"}
        mapping = {1: "no", 4: "no", 5: "no", 6: "no"}
    else:
        mapping = {}

    return mapping.get(rid, "no")
```

Function print :

1.2 Metadata

Nombre de lignes df

7932

Nombre de lignes df clean	7932
Nombre de features	32
% de NaN	0.0
Taille dataset entraînement	5552
Taille dataset test	2379
Métrique utilisée	r2

Column target : i_NH3

Features_X : r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_wl_6h_D1_energy, r_NH3_wl_6h_D2_energy, r_NH3_wl_6h_D3_energy, r_NH3_wl_6h_A_energy, r_NH3_wl_6h_A_rms, r_NH3_wl_6h_D1_energy_rel, r_NH3_wl_6h_D2_energy_rel, r_NH3_wl_6h_D3_energy_rel, r_NH3_wl_6h_A_energy_rel, r_temperature_wl_6h_D1_energy, r_temperature_wl_6h_D2_energy, r_temperature_wl_6h_D3_energy, r_temperature_wl_6h_A_energy, r_temperature_wl_6h_A_rms, r_temperature_wl_6h_D1_energy_rel, r_temperature_wl_6h_D2_energy_rel, r_temperature_wl_6h_D3_energy_rel, r_temperature_wl_6h_A_energy_rel, r_umidity_wl_6h_D1_energy, r_umidity_wl_6h_D2_energy, r_umidity_wl_6h_D3_energy, r_umidity_wl_6h_A_energy, r_umidity_wl_6h_A_rms, r_umidity_wl_6h_D1_energy_rel, r_umidity_wl_6h_D2_energy_rel, r_umidity_wl_6h_D3_energy_rel, r_umidity_wl_6h_A_energy_rel

feats_to_balance :

feats_grouping :

feats_postproc : datetime, id_channel, id_rabbit, campaign

feats_ref :

id_cols : campaign

2 Data correlation and clustering

2.1 Clustermap and cluster list

The following section has been done with a threshold choosen of : 0.8

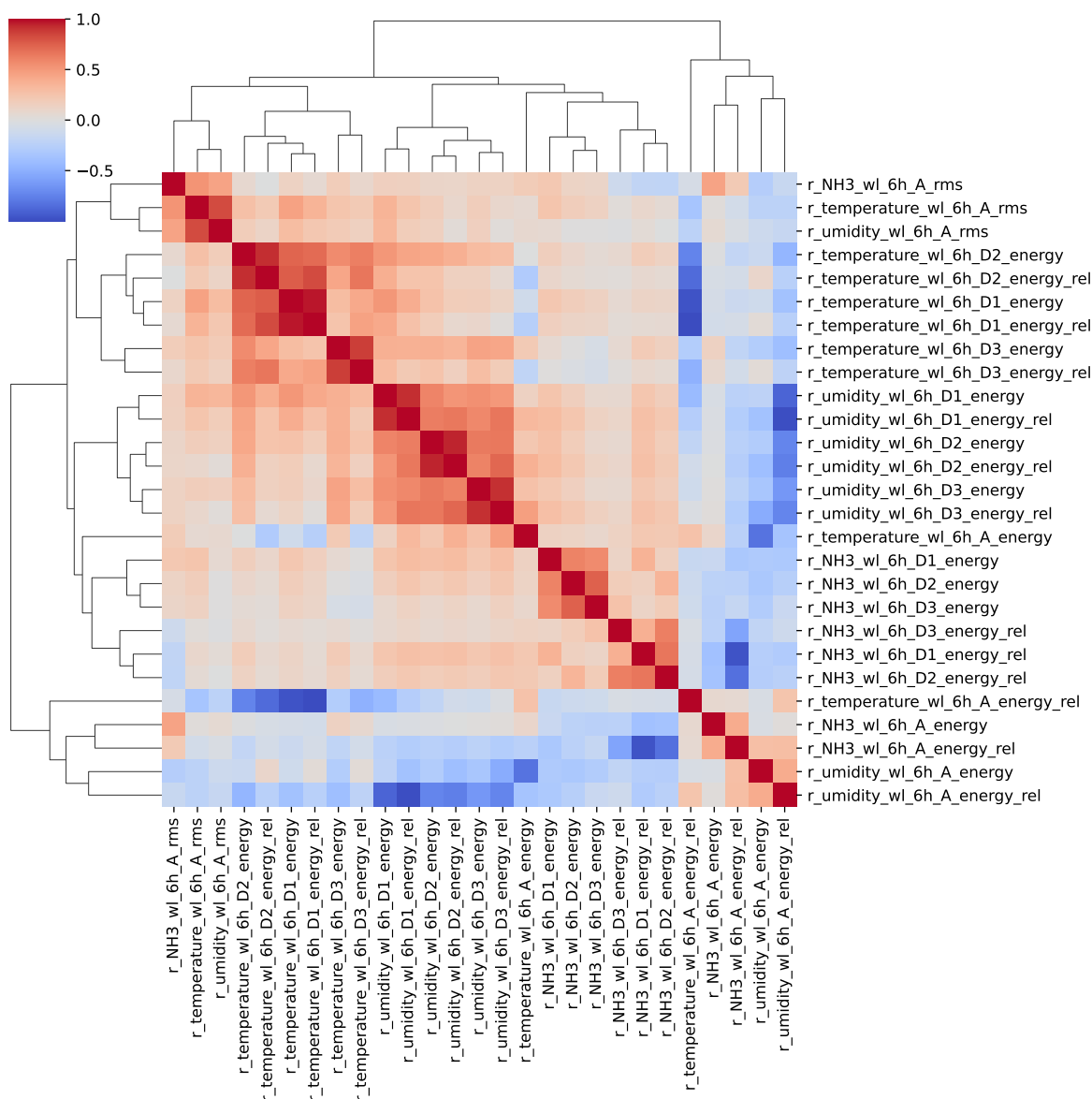
cluster 1: r, , N, H, 3 ; **cluster 2:** r, , t, e, m, p, e, r, a, t, u, r, e ; **cluster 3:** r, , u, m, i, d, i, t, y ; **cluster 4:** r, , C, O, 2 ; **cluster 5:** r, , T, H, I ; **cluster 6:** r_NH3_wl_6h_D1_energy ; **cluster 7:** r_NH3_wl_6h_D2_energy ; **cluster 8:** r_NH3_wl_6h_D3_energy ; **cluster 9:** r_NH3_wl_6h_A_energy ; **cluster 10:** r_NH3_wl_6h_A_rms ; **cluster 11:**

r_NH3_wl_6h_D2_energy_rel, r_NH3_wl_6h_D1_energy_rel, r_NH3_wl_6h_A_energy_rel ; **cluster 12:** r_NH3_wl_6h_D3_energy_rel ; **cluster 13:** r_temperature_wl_6h_D2_energy, r_temperature_wl_6h_D1_energy, r_temperature_wl_6h_A_energy_rel, r_temperature_wl_6h_D1_energy_rel, r_temperature_wl_6h_D2_energy_rel ; **cluster 14:** r_temperature_wl_6h_D3_energy, r_temperature_wl_6h_D3_energy_rel ; **cluster 15:** r_temperature_wl_6h_A_energy, r_umidity_wl_6h_A_energy ; **cluster 16:** r_temperature_wl_6h_A_rms, r_umidity_wl_6h_A_rms ; **cluster 17:** r_umidity_wl_6h_D1_energy, r_umidity_wl_6h_D1_energy_rel, r_umidity_wl_6h_A_energy_rel ; **cluster 18:** r_umidity_wl_6h_D2_energy, r_umidity_wl_6h_D2_energy_rel ; **cluster 19:** r_umidity_wl_6h_D3_energy, r_umidity_wl_6h_D3_energy_rel ;

2.2 Features selection

Only the first items of the clusters is choosen for the rest of the study

Choosen features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_wl_6h_D1_energy, r_NH3_wl_6h_D2_energy, r_NH3_wl_6h_D3_energy, r_NH3_wl_6h_A_energy, r_NH3_wl_6h_A_rms, r_NH3_wl_6h_D2_energy_rel, r_NH3_wl_6h_D3_energy_rel, r_temperature_wl_6h_D2_energy, r_temperature_wl_6h_D3_energy, r_temperature_wl_6h_A_energy, r_temperature_wl_6h_A_rms, r_umidity_wl_6h_D1_energy, r_umidity_wl_6h_D2_energy, r_umidity_wl_6h_D3_energy



3 Model(s) analysis

3.1 Metadata

Features X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_wl_6h_D1_energy, r_NH3_wl_6h_D2_energy, r_NH3_wl_6h_D3_energy, r_NH3_wl_6h_A_energy, r_NH3_wl_6h_A_rms, r_NH3_wl_6h_D2_energy_rel, r_NH3_wl_6h_D3_energy_rel,

r_temperature_wl_6h_D2_energy, r_temperature_wl_6h_D3_energy, r_temperature_wl_6h_A_energy, r_temperature_wl_6h_A_rms, r_umidity_wl_6h_D1_energy, r_umidity_wl_6h_D2_energy, r_umidity_wl_6h_D3_energy

Len(Features_X): 19

Config Gridsearch:{‘show’: True, ‘search_type’: ‘optuna’, ‘model_name’: None, ‘model’: None, ‘param_grid’: None, ‘best_param’: None, ‘cv_func’: ShuffleSplit(n_splits=5, random_state=42, test_size=0.3, train_size=None), ‘save_models’: True, ‘sumup-table’: True}

3.2 Explanation legend tables

Columns

- rank: ranking regarding cross validation test
- train_cv: mean score train of the CV
- test_cv: mean score test of the CV
- test: score on the testing dataset
- diff: test-test_cv
- r_test: test/test_cv
- r_train: train_cv/test_cv

Colors:

[Orange] $r_test < 0.95$ (under performance of the test)

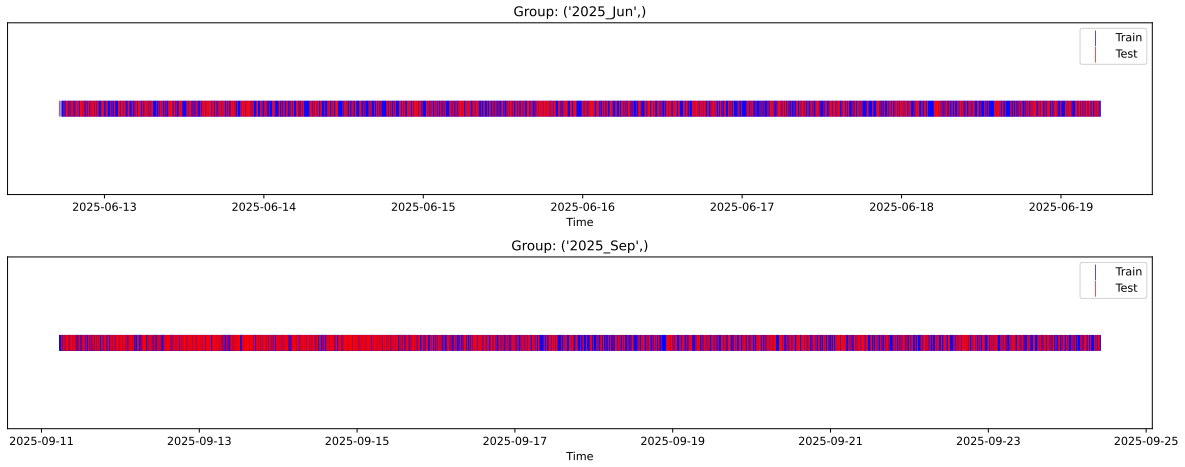
[Green] $r_test > 1.05$ (over performance of the test)

[Red] r_train not in $[0.95, 1.05]$ (gap between train/test)

3.3 Repartition of train and test

3.3.1 External Split: X_train vs X_test

External Split: Train: 5552 samples Test: 2380 samples



3.3.2 Internal CV Folds

Fold 1/5

Train: 3886 samples Test: 1666 samples Excluded: 0 samples Window exclusion: 0h



Fold 2/5

Train: 3886 samples Test: 1666 samples Excluded: 0 samples Window exclusion: 0h



Fold 3/5

Train: 3886 samples Test: 1666 samples Excluded: 0 samples Window exclusion: 0h



Fold 4/5

Train: 3886 samples Test: 1666 samples Excluded: 0 samples Window exclusion: 0h



Fold 5/5

Train: 3886 samples Test: 1666 samples Excluded: 0 samples Window exclusion: 0h



3.4 Model : Linear

3.4.1 Gridsearch

Distribution y_{train} vs y_{test} : y_{train} : mean=2.357, std=0.808, min=0.959, max=13.845
 y_{test} : mean=2.347, std=0.779, min=0.973, max=8.190

Fold 0 — y_{test} : mean=2.36, std=0.82, y_{train} : mean=2.36, std=0.80

Fold 1 — y_{test} : mean=2.36, std=0.79, y_{train} : mean=2.36, std=0.81

Fold 2 — y_{test} : mean=2.38, std=0.80, y_{train} : mean=2.35, std=0.81

Fold 3 — y_test: mean=2.36, std=0.85, y_train: mean=2.36, std=0.79

Fold 4 — y_test: mean=2.35, std=0.82, y_train: mean=2.36, std=0.80

train scores CV: [0.53454089 0.4969492 0.50213929 0.53609424 0.53948389]

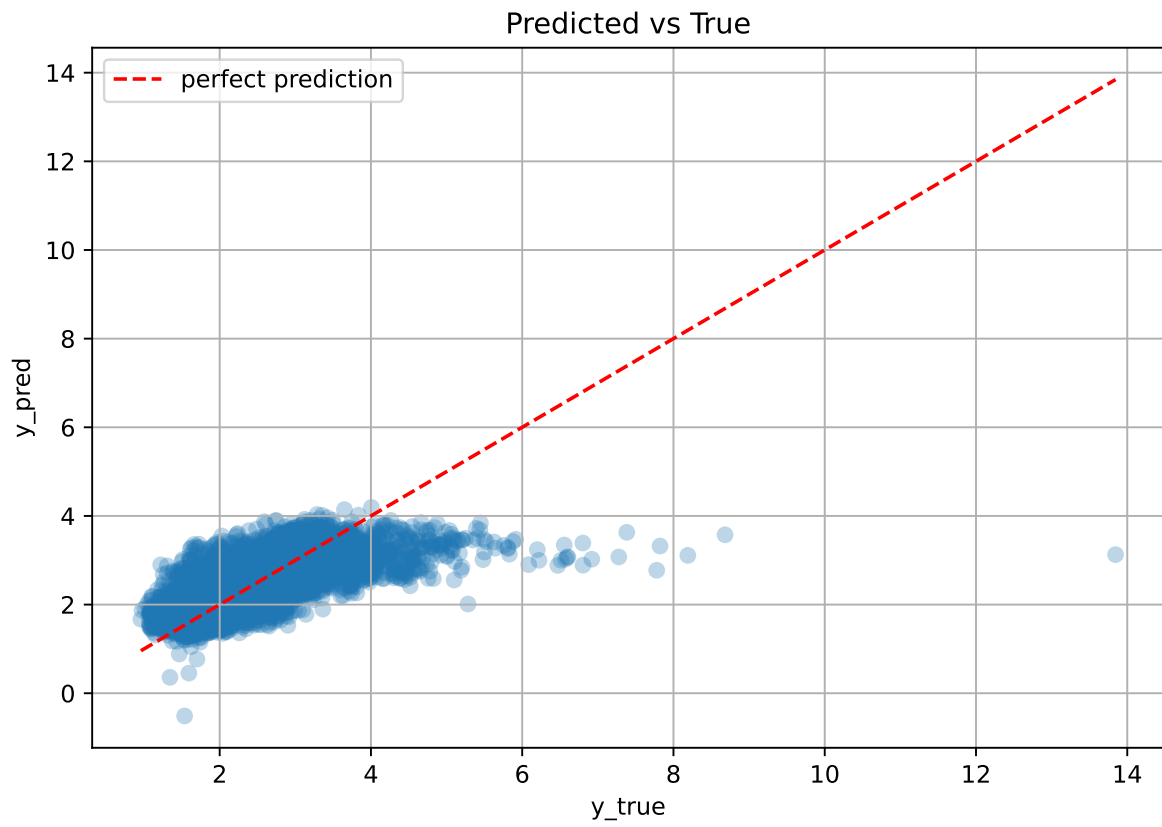
test scores CV: [0.35712926 0.54668446 0.53292092 0.39792294 0.44760755]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.5218	0.4565	0.074	0.5374	0.5373	0.0808	1.1771	1.1433

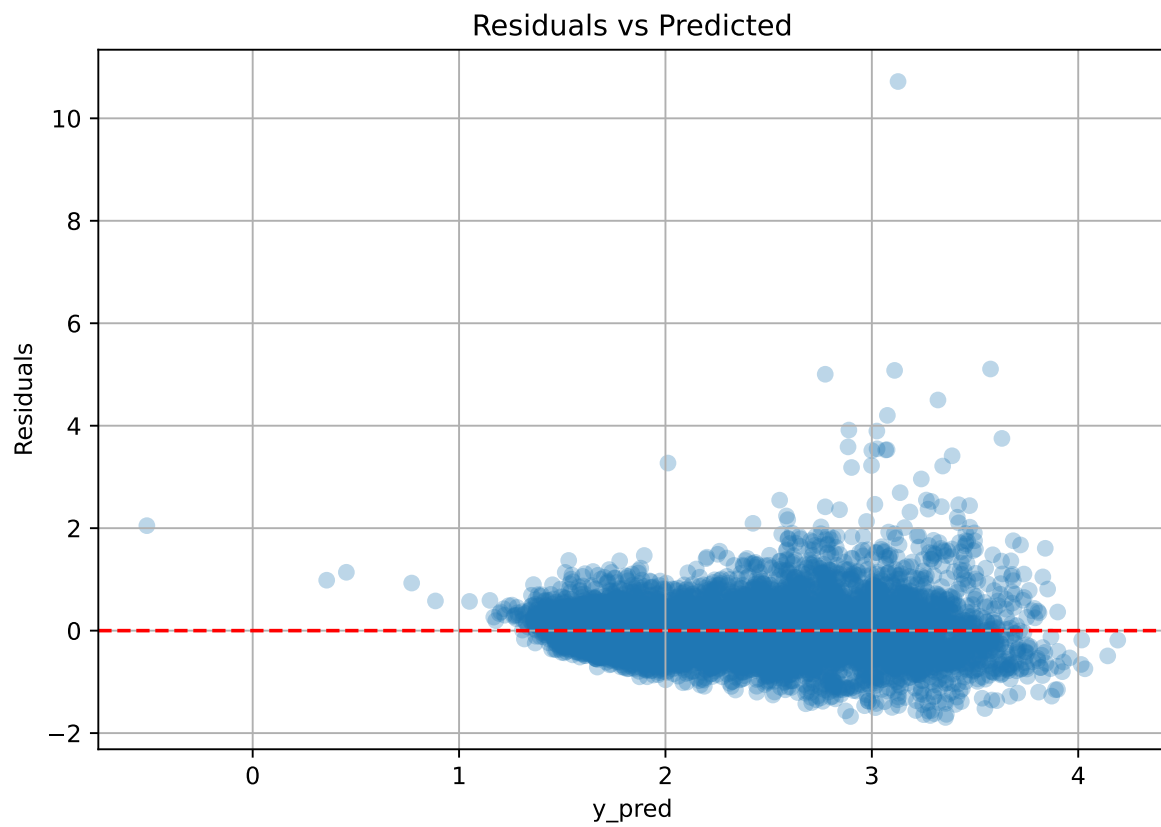
Coefficients:

coef_r_CO2 = -0.0580, coef_r_NH3 = 0.2358, coef_r_NH3_wl_6h_A_energy = -0.2810, coef_r_NH3_wl_6h_A_rms = -0.0248, coef_r_NH3_wl_6h_D1_energy = 0.0475, coef_r_NH3_wl_6h_D2_energy = 0.1204, coef_r_NH3_wl_6h_D2_energy_rel = -0.1101, coef_r_NH3_wl_6h_D3_energy = -0.0770, coef_r_NH3_wl_6h_D3_energy_rel = 0.0067, coef_r_THI = 1.2013, coef_r_temperature = -1.1056, coef_r_temperature_wl_6h_A_energy = -0.3499, coef_r_temperature_wl_6h_A_rms = -0.0416, coef_r_temperature_wl_6h_D2_energy = 0.0749, coef_r_temperature_wl_6h_D3_energy = -0.1233, coef_r_umidity = -0.7620, coef_r_umidity_wl_6h_D1_energy = -0.0075, coef_r_umidity_wl_6h_D2_energy = -0.0232, coef_r_umidity_wl_6h_D3_energy = 0.0193, intercept = 2.3570,

3.4.2 Scatter



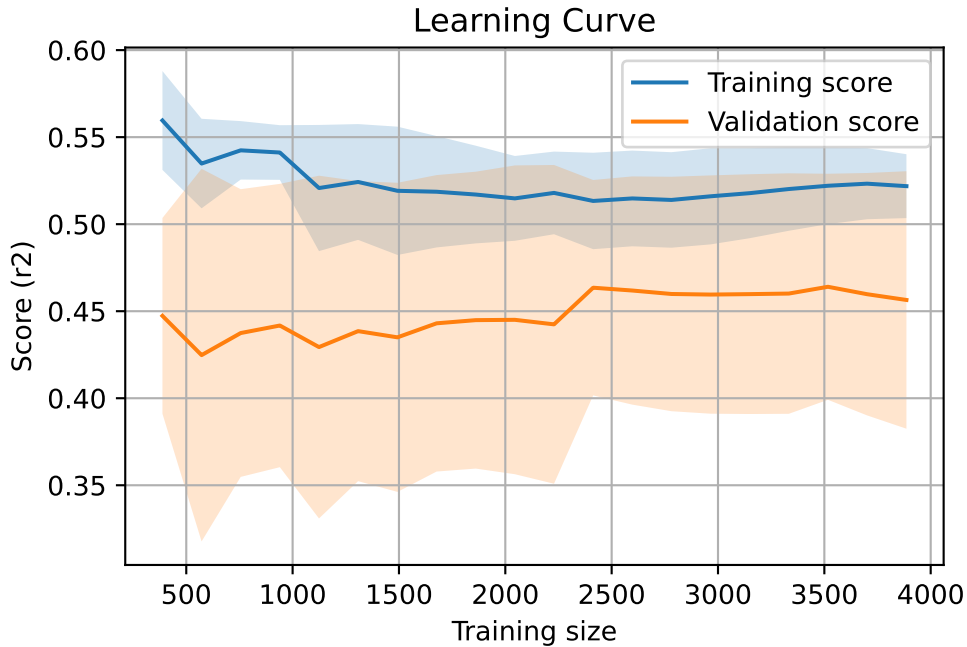
3.4.3 Scatter residuals



3.4.4 Learning curve

`Len(group)` : 7932, `Len(training)` : 5553,

`Len(training_real)` : 3888, `Len(test_of_training)` : 1665, `Len(test)` : 2379



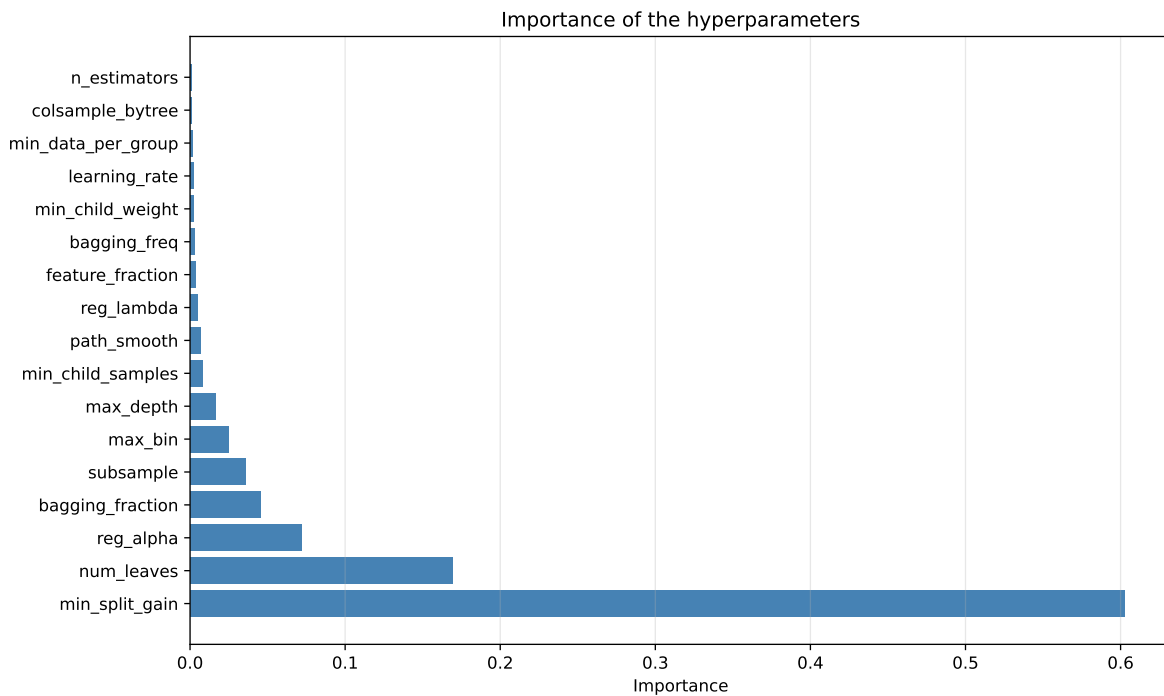
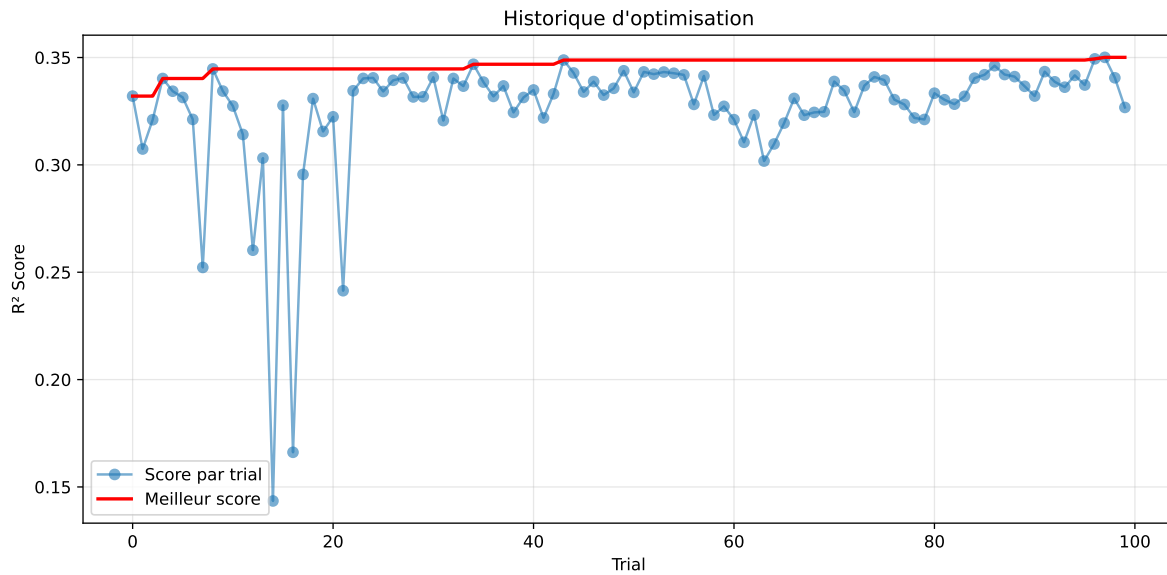
3.5 Model : LGBM

3.5.1 Gridsearch

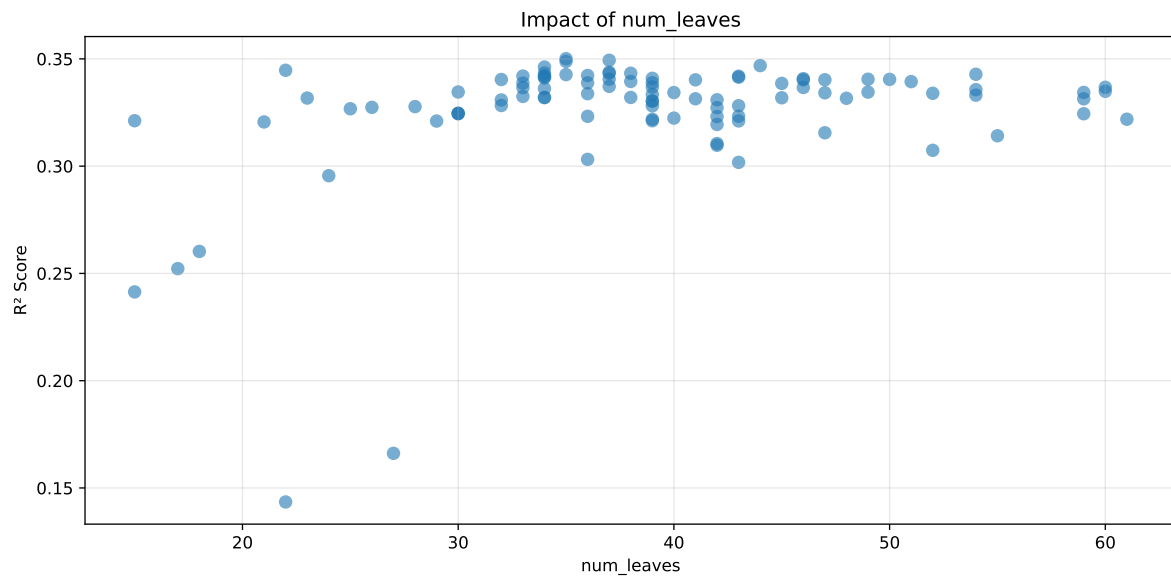
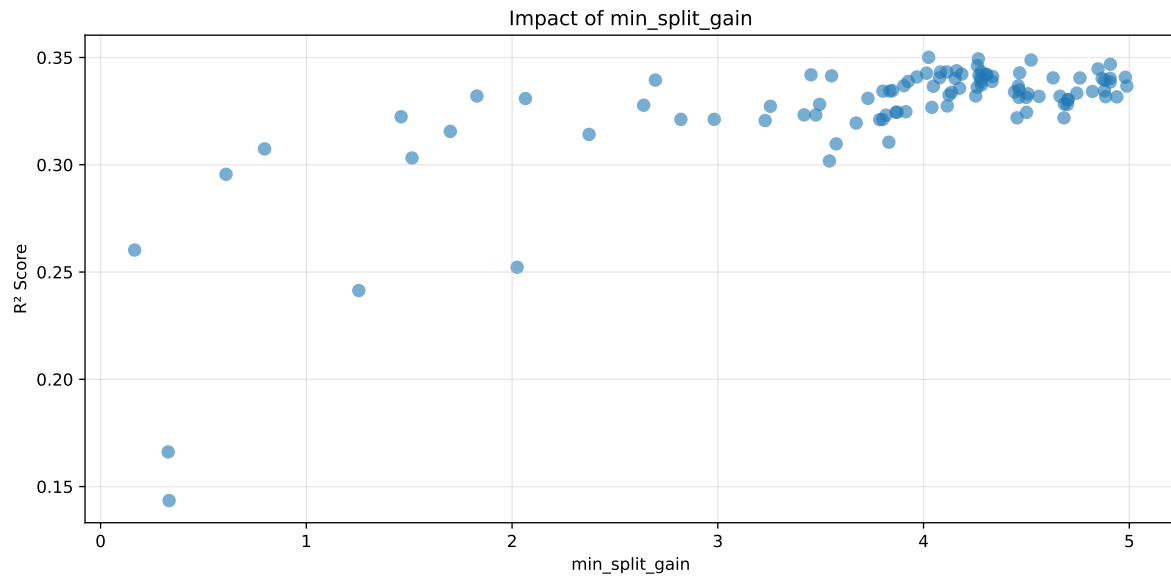
Distribution y_{train} vs y_{test} : y_{train} : mean=2.357, std=0.808, min=0.959, max=13.845
 y_{test} : mean=2.347, std=0.779, min=0.973, max=8.190

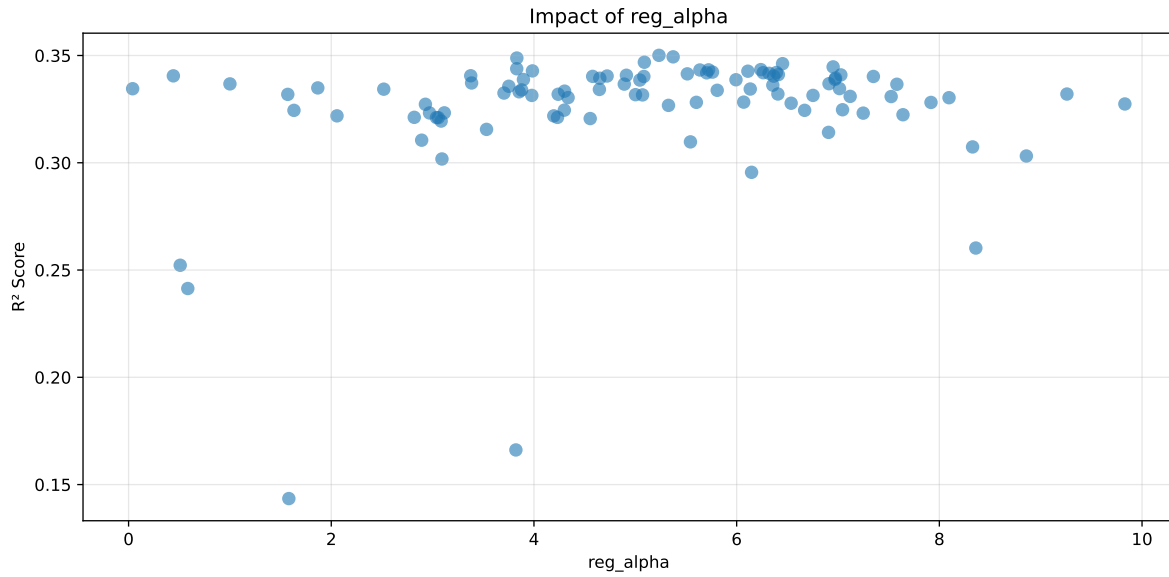
===== TOP Importance FEATURES =====

feature	importance
r_temperature_wl_6h_A_energy	49
r_umidity	26
r_temperature	26
r_CO2	25
r_NH3_wl_6h_D1_energy	22
r_temperature_wl_6h_D2_energy	16
r_NH3_wl_6h_A_energy	15
r_NH3_wl_6h_D3_energy_rel	15
r_NH3_wl_6h_D2_energy	14
r_NH3	11
r_THI	11
r_umidity_wl_6h_D1_energy	10
r_NH3_wl_6h_D3_energy	8
r_temperature_wl_6h_A_rms	7
r_umidity_wl_6h_D2_energy	6
r_NH3_wl_6h_A_rms	4
r_NH3_wl_6h_D2_energy_rel	4
r_temperature_wl_6h_D3_energy	2
r_umidity_wl_6h_D3_energy	0



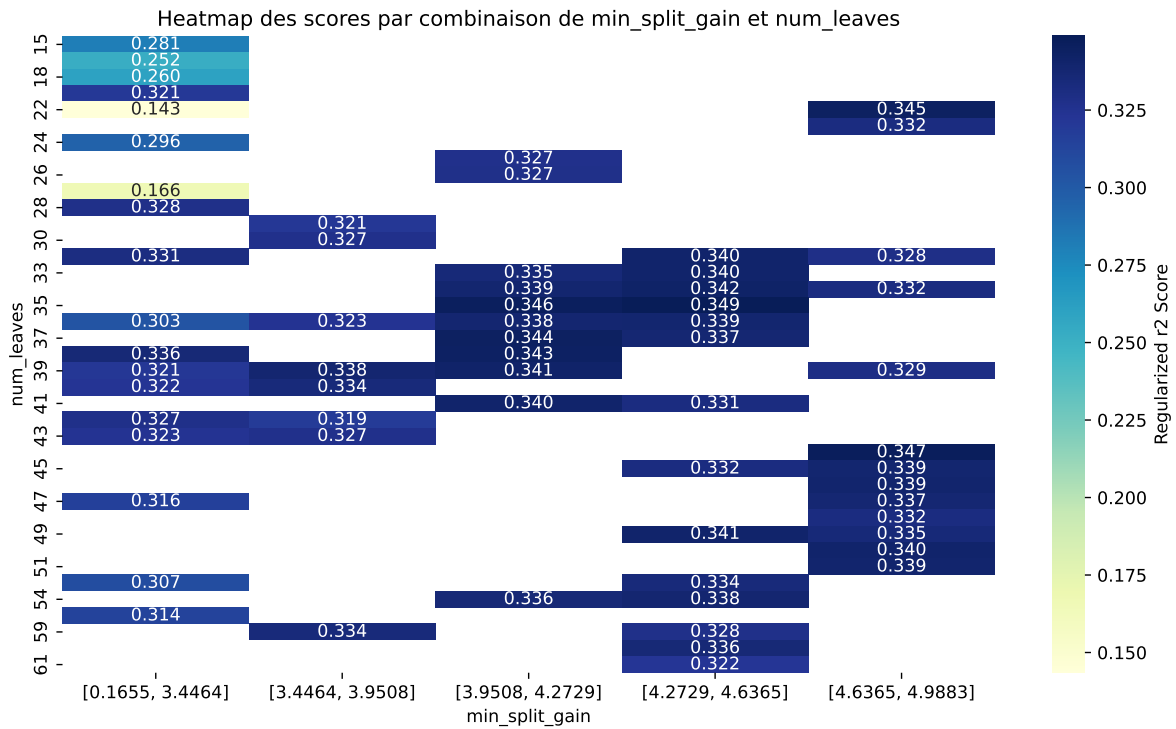
Paramètres avec >5% d'importance : ['min_split_gain', 'num_leaves', 'reg_alpha']





C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.6243	0.5786	0.6145	0.0421	0.3501	0.6355	0.0569	1.0983	1.079
2	0.6225	0.5769	0.6115	0.0402	0.3494	0.6344	0.0574	1.0995	1.0789
3	0.6242	0.5783	0.6135	0.0416	0.3488	0.63	0.0517	1.0893	1.0794
4	0.6142	0.5696	0.6053	0.0442	0.3469	0.6249	0.0552	1.0969	1.0782
5	0.595	0.5535	0.5894	0.0413	0.3462	0.6142	0.0607	1.1097	1.0749
6	0.5943	0.5527	0.5873	0.0392	0.3447	0.615	0.0624	1.1129	1.0753
7	0.6209	0.5748	0.6102	0.0405	0.3438	0.6379	0.0632	1.1099	1.0804
8	0.5934	0.5517	0.5889	0.0432	0.3434	0.6016	0.0499	1.0904	1.0755
9	0.6207	0.5745	0.6093	0.0438	0.3433	0.6365	0.062	1.108	1.0805
10	0.6151	0.5698	0.6061	0.044	0.3432	0.6315	0.0618	1.1084	1.0795
---	---	---	---	---	---	---	---	---	---
100	0.7993	0.69	0.7401	0.038	0.1435	0.7587	0.0687	1.0996	1.1584

Hyperparameters:

Rank 1: {'reg__n_estimators': 300, 'reg__max_depth': 3, 'reg__learning_rate': 0.07692872867546793, 'reg__num_leaves': 35, 'reg__subsample': 0.6094274486114082, 'reg__colsample_bytree': 0.7631883366869814, 'reg__min_child_samples': 42, 'reg__reg_alpha': 5.233290989912659, 'reg__reg_lambda': 10.142669741658192, 'reg__min_split_gain': 4.024736437752662, 'reg__path_smooth': 1.531735550928702, 'reg__min_child_weight': 0.02802593186308257, 'reg__feature_fraction': 0.5521427043513185, 'reg__bagging_fraction': 0.6850795566241076, 'reg__bagging_freq': 3, 'reg__max_bin': 69, 'reg__min_data_per_group': 116}

Rank 2: {'reg__n_estimators': 300, 'reg__max_depth': 3, 'reg__learning_rate': 0.07564428506130223, 'reg__num_leaves': 37, 'reg__subsample': 0.6130326523269317, 'reg__colsample_bytree': 0.7576375881043543, 'reg__min_child_samples': 37, 'reg__reg_alpha': 5.37325416457792, 'reg__reg_lambda': 7.469973350328743, 'reg__min_split_gain': 4.266398006053841, 'reg__path_smooth': 2.9963976939806254, 'reg__min_child_weight': 0.02895820166369849, 'reg__feature_fraction': 0.5483138516489574, 'reg__bagging_fraction': 0.6893491157698268, 'reg__bagging_freq': 3, 'reg__max_bin': 73, 'reg__min_data_per_group': 113}

Rank 3: {'reg__n_estimators': 700, 'reg__max_depth': 4, 'reg__learning_rate': 0.04489079101736438, 'reg__num_leaves': 35, 'reg__subsample': 0.6265039445709982, 'reg__colsample_bytree': 0.7221075109393424, 'reg__min_child_samples': 32, 'reg__reg_alpha': 3.83098270044253, 'reg__reg_lambda': 6.050569722938685, 'reg__min_split_gain': 4.522548104637166, 'reg__path_smooth': 3.7137018782472078, 'reg__min_child_weight': 0.17845999435918536, 'reg__feature_fraction': 0.5879248084542958, 'reg__bagging_fraction': 0.5533002539466096, 'reg__bagging_freq': 2, 'reg__max_bin': 90, 'reg__min_data_per_group': 168}

Rank 4: {'reg__n_estimators': 500, 'reg__max_depth': 3, 'reg__learning_rate': 0.07772163019998642, 'reg__num_leaves': 44, 'reg__subsample': 0.5711311314942181, 'reg__colsample_bytree': 0.7377768889270934, 'reg__min_child_samples': 40, 'reg__reg_alpha': 5.08929550879227, 'reg__reg_lambda': 8.010918471779828, 'reg__min_split_gain': 4.907885745701114, 'reg__path_smooth': 0.04368068993591945, 'reg__min_child_weight': 0.0061888118074936126, 'reg__feature_fraction': 0.5862718250186137, 'reg__bagging_fraction': 0.6434403457721725, 'reg__bagging_freq': 2, 'reg__max_bin': 63, 'reg__min_data_per_group': 198}

Rank 5: {'reg__n_estimators': 300, 'reg__max_depth': 3, 'reg__learning_rate': 0.03287533444858392, 'reg__num_leaves': 34, 'reg__subsample': 0.6151280040245527, 'reg__colsample_bytree': 0.8290628599826428, 'reg__min_child_samples': 40, 'reg__reg_alpha': 6.452995342542708, 'reg__reg_lambda': 17.175795129869858, 'reg__min_split_gain': 4.261899916422554, 'reg__path_smooth': 2.6944836328360826, 'reg__min_child_weight': 0.32893429312146216, 'reg__feature_fraction': 0.5523915052825108, 'reg__bagging_fraction': 0.5148799776285806, 'reg__bagging_freq': 7, 'reg__max_bin': 71, 'reg__min_data_per_group': 194}

Rank 6: {'reg__n_estimators': 400, 'reg__max_depth': 4, 'reg__learning_rate': 0.07196422577772463, 'reg__num_leaves': 22, 'reg__subsample': 0.6330635858224839, 'reg__colsample_bytree': 0.8134767597834567, 'reg__min_child_samples': 38, 'reg__reg_alpha': 6.9518621251390265, 'reg__reg_lambda': 14.61802788234437, 'reg__min_split_gain': 4.848047051098392, 'reg__path_smooth': 3.04737770785912, 'reg__min_child_weight': 4.790599969580461, 'reg__feature_fraction': 0.6094227805781731, 'reg__bagging_fraction': 0.5250713454591648, 'reg__bagging_freq': 4, 'reg__max_bin': 65, 'reg__min_data_per_group': 193}

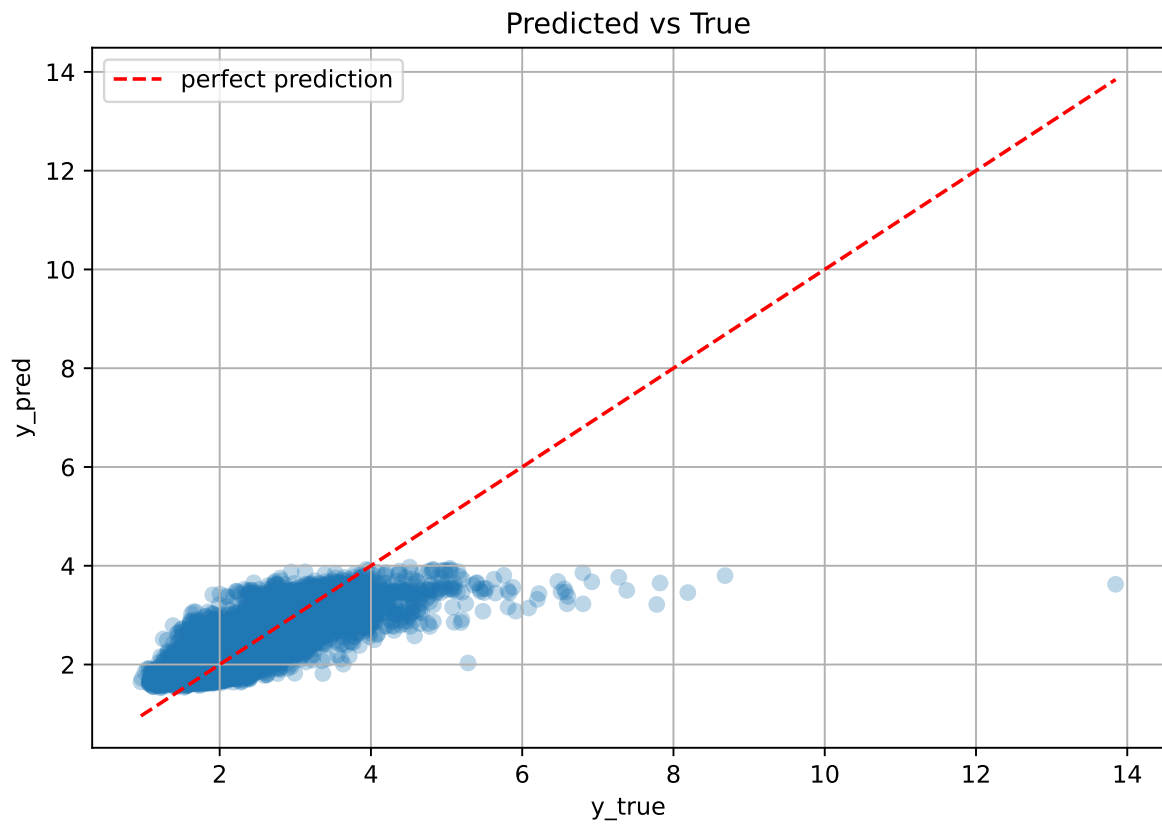
Rank 7: {'reg__n_estimators': 300, 'reg__max_depth': 3, 'reg__learning_rate': 0.09161306224951635, 'reg__num_leaves': 37, 'reg__subsample': 0.6426432284383349, 'reg__colsample_bytree': 0.8427936077992463, 'reg__min_child_samples': 38, 'reg__reg_alpha': 3.82999559664698, 'reg__reg_lambda': 5.415149336258617, 'reg__min_split_gain': 4.160848218931685, 'reg__path_smooth': 0.046836810287085306, 'reg__min_child_weight': 1.3398053906755618, 'reg__feature_fraction': 0.5920307662139349, 'reg__bagging_fraction': 0.5591549983514063, 'reg__bagging_freq': 5, 'reg__max_bin': 81, 'reg__min_data_per_group': 110}

Rank 8: {'reg__n_estimators': 300, 'reg__max_depth': 3, 'reg__learning_rate': 0.04853749669117618, 'reg__num_leaves': 34, 'reg__subsample': 0.6133386682174541, 'reg__colsample_bytree': 0.7510437817704226, 'reg__min_child_samples': 40, 'reg__reg_alpha': 6.238462969469989, 'reg__reg_lambda': 48.44735994866063, 'reg__min_split_gain': 4.280302834348563, 'reg__path_smooth': 2.7393346841814123, 'reg__min_child_weight': 0.09164876705754696, 'reg__feature_fraction': 0.5538536630523025, 'reg__bagging_fraction': 0.5116285826440037, 'reg__bagging_freq': 3, 'reg__max_bin': 73, 'reg__min_data_per_group': 73}

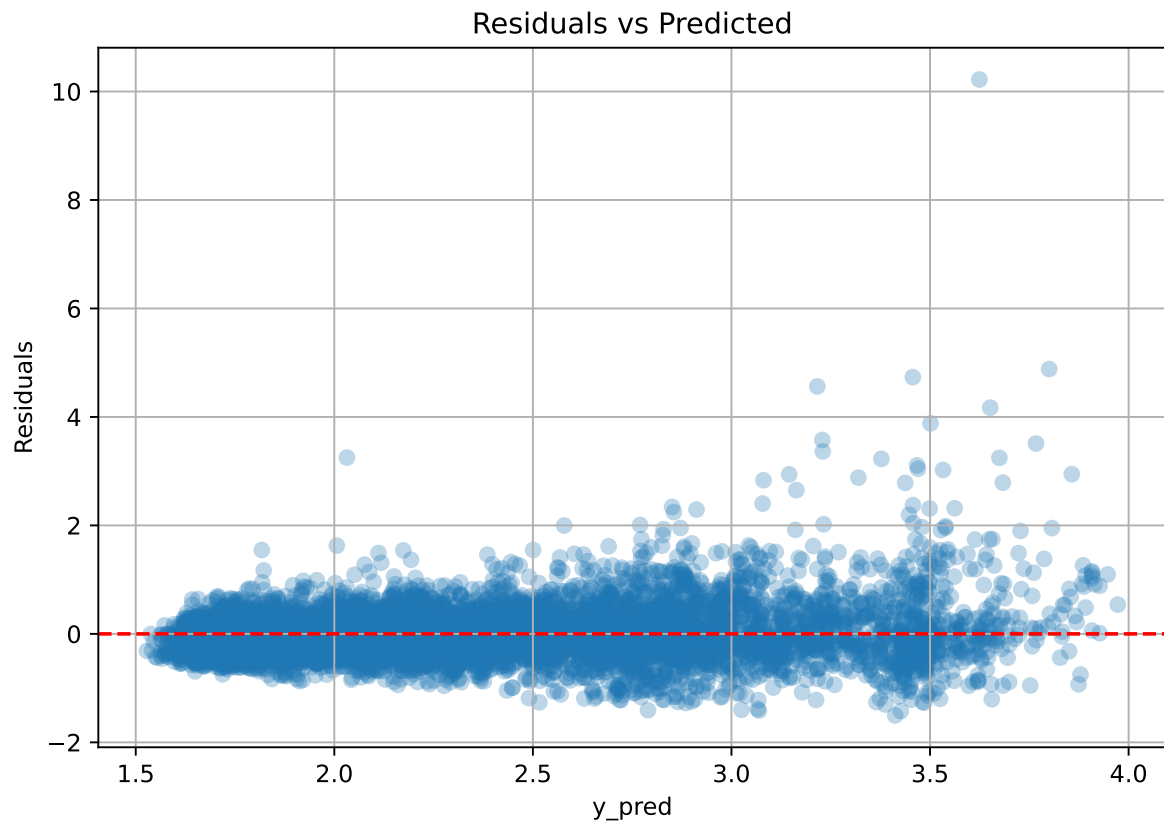
Rank 9: {'reg__n_estimators': 300, 'reg__max_depth': 4, 'reg__learning_rate': 0.08968423808355945, 'reg__num_leaves': 38, 'reg__subsample': 0.646199834899952, 'reg__colsample_bytree': 0.8477597316881699, 'reg__min_child_samples': 34, 'reg__reg_alpha': 5.723301019025052, 'reg__reg_lambda': 4.458633078425539, 'reg__min_split_gain': 4.113333999103841, 'reg__path_smooth': 3.7859244935210272, 'reg__min_child_weight': 0.15212023745721492, 'reg__feature_fraction': 0.5869867138723317, 'reg__bagging_fraction': 0.5552564008703076, 'reg__bagging_freq': 5, 'reg__max_bin': 108, 'reg__min_data_per_group': 173}

Rank 10: {'reg__n_estimators': 300, 'reg__max_depth': 3, 'reg__learning_rate': 0.09206707879168627, 'reg__num_leaves': 37, 'reg__subsample': 0.6501052598789517, 'reg__colsample_bytree': 0.8609047795348845, 'reg__min_child_samples': 38, 'reg__reg_alpha': 5.637308242047176, 'reg__reg_lambda': 5.45009059997067, 'reg__min_split_gain': 4.081989512407003, 'reg__path_smooth': 1.0180851072930008, 'reg__min_child_weight': 0.12302716979474651, 'reg__feature_fraction': 0.5961233131907968, 'reg__bagging_fraction': 0.5619058462688765, 'reg__bagging_freq': 3, 'reg__max_bin': 80, 'reg__min_data_per_group': 175}

3.5.2 Scatter



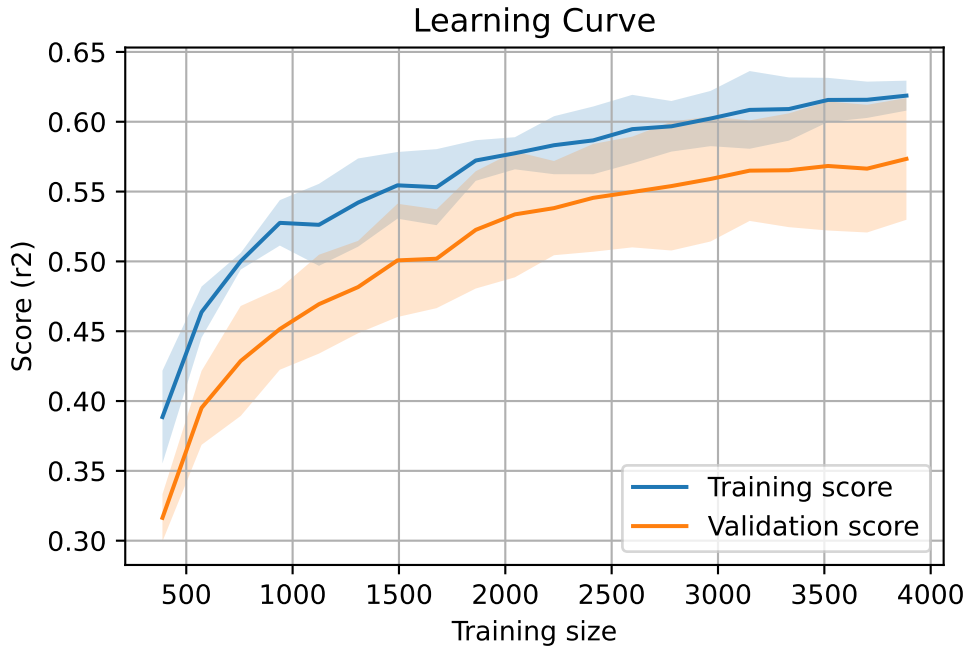
3.5.3 Scatter residuals



3.5.4 Learning curve

`Len(group)` : 7932, `Len(training)` : 5553,

`Len(training_real)` : 3888, `Len(test_of_training)` : 1665, `Len(test)` : 2379



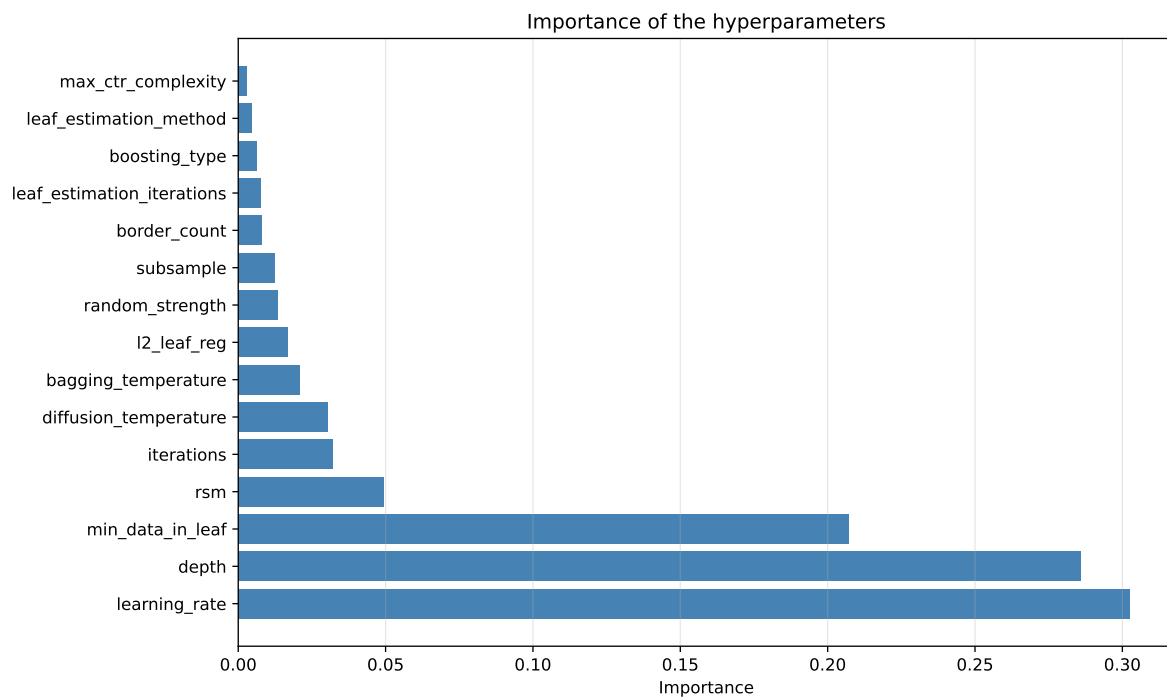
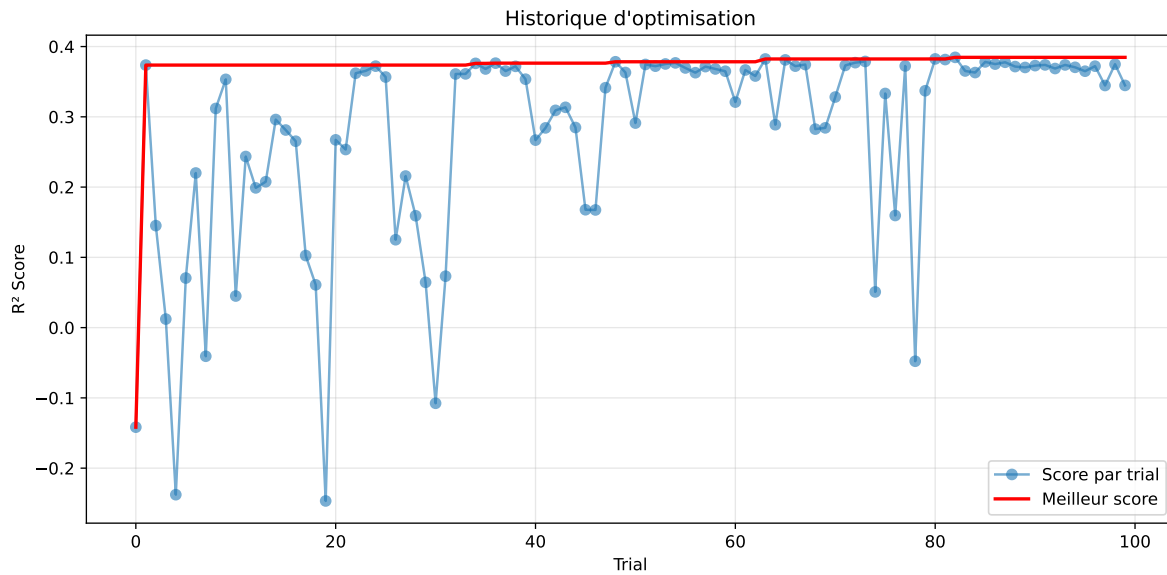
3.6 Model : CatBoost

3.6.1 Gridsearch

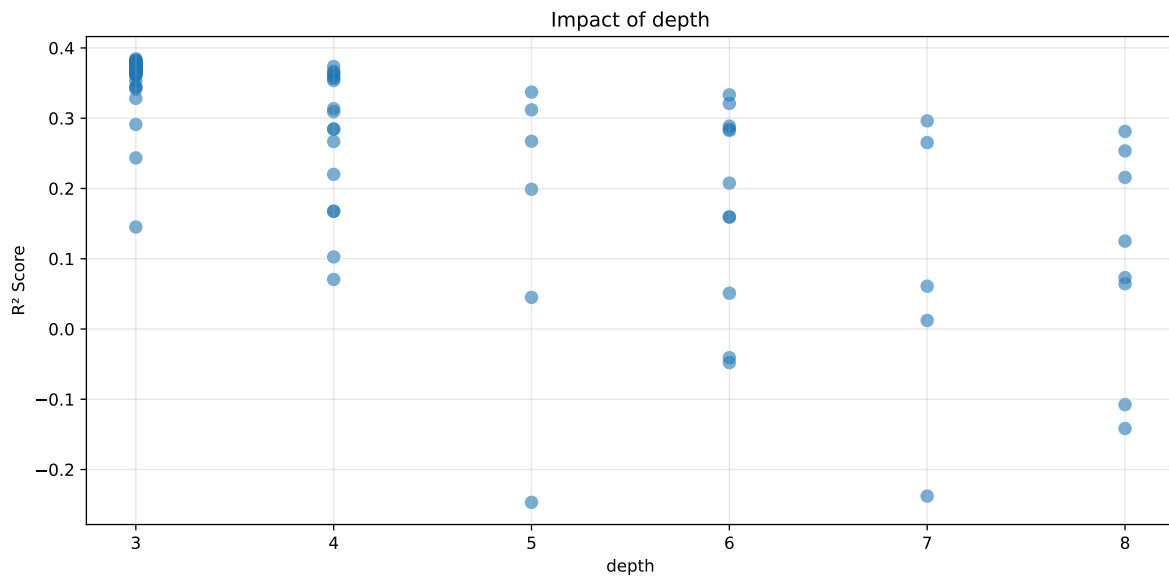
Distribution y_train vs y_test: y_train: mean=2.357, std=0.808, min=0.959, max=13.845
y_test: mean=2.347, std=0.779, min=0.973, max=8.190

===== TOP Importance FEATURES =====

feature	importance
r_umidity	24.472471
r_temperature_wl_6h_A_energy	21.889588
r_temperature	15.430719
r_temperature_wl_6h_D2_energy	5.627033
r_THI	4.932425
r_NH3_wl_6h_D1_energy	3.984092
r_umidity_wl_6h_D1_energy	3.882005
r_CO2	3.289026
r_NH3	3.080604
r_NH3_wl_6h_D2_energy	2.792269
r_NH3_wl_6h_D3_energy	2.140618
r_NH3_wl_6h_A_energy	2.033698
r_umidity_wl_6h_D2_energy	1.494400
r_temperature_wl_6h_A_rms	1.440483
r_NH3_wl_6h_D3_energy_rel	1.216092
r_NH3_wl_6h_A_rms	0.935468
r_temperature_wl_6h_D3_energy	0.586588
r_NH3_wl_6h_D2_energy_rel	0.504865
r_umidity_wl_6h_D3_energy	0.267557



Paramètres avec >5% d'importance : ['learning_rate', 'depth', 'min_data_in_leaf']



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.6312	0.5901	0.6296	0.0424	0.3846	0.636	0.0459	1.0778	1.0696
2	0.6271	0.5864	0.6258	0.0409	0.3824	0.6322	0.0458	1.0781	1.0696
3	0.6532	0.608	0.648	0.042	0.3823	0.6542	0.0462	1.0759	1.0742
4	0.6418	0.5985	0.6382	0.0417	0.3815	0.6421	0.0436	1.0729	1.0725
5	0.6531	0.6077	0.6477	0.0426	0.381	0.6518	0.044	1.0724	1.0746
6	0.6486	0.6037	0.6428	0.0419	0.3788	0.6496	0.0459	1.0761	1.0745
7	0.6723	0.6233	0.6632	0.0428	0.3784	0.6705	0.0472	1.0758	1.0786
8	0.626	0.5847	0.623	0.043	0.3781	0.6265	0.0418	1.0714	1.0707
9	0.6254	0.5841	0.6239	0.0406	0.3776	0.6299	0.0458	1.0784	1.0707
10	0.6541	0.6079	0.6488	0.0415	0.3771	0.6524	0.0445	1.0733	1.0759
---	---	---	---	---	---	---	---	---	---
100	0.9285	0.7326	0.7821	0.0367	-0.2467	0.795	0.0624	1.0851	1.2673

Hyperparameters:

Rank 1: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.023180826374366057, 'reg__l2_leaf_reg': 19.002757731434333, 'reg__bagging_temperature': 3.388694462943975, 'reg__random_strength': 2.765125336268173, 'reg__subsample': 0.7701493011237738, 'reg__min_data_in_leaf': 21, 'reg__leaf_estimation_iterations': 8, 'reg__rsm': 0.6649242451060168, 'reg__border_count': 38, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 2516.092177591128}

Rank 2: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.022002545081961526, 'reg__l2_leaf_reg': 22.98449528437498, 'reg__bagging_temperature': 2.9766490309374083, 'reg__random_strength': 2.722801394199669, 'reg__subsample': 0.7605024122584233, 'reg__min_data_in_leaf': 23, 'reg__leaf_estimation_iterations': 8, 'reg__rsm': 0.5716421184643069, 'reg__border_count': 47, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 2957.301183065976}

Rank 3: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.02749632430336207, 'reg__l2_leaf_reg': 20.00704679188065, 'reg__bagging_temperature': 2.7222269116720548, 'reg__random_strength': 2.2917006564903257, 'reg__subsample': 0.786400832200342, 'reg__min_data_in_leaf': 17, 'reg__leaf_estimation_iterations': 6, 'reg__rsm': 0.6369461814052069, 'reg__border_count': 39, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 4555.49095223681}

Rank 4: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.022283186111873007, 'reg__l2_leaf_reg': 2.433303657244103, 'reg__bagging_temperature': 3.4031288892860525, 'reg__random_strength': 1.6215928920266878, 'reg__subsample': 0.766783523740397, 'reg__min_data_in_leaf': 21, 'reg__leaf_estimation_iterations': 8, 'reg__rsm':

0.5607583355396115, 'reg__border_count': 38, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 2932.798879799723}

Rank 5: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.0278329296858231, 'reg__l2_leaf_reg': 21.00009390487394, 'reg__bagging_temperature': 2.8618584236596885, 'reg__random_strength': 2.251145686490343, 'reg__subsample': 0.7816034413529535, 'reg__min_data_in_leaf': 18, 'reg__leaf_estimation_iterations': 6, 'reg__rsm': 0.6419789855693865, 'reg__border_count': 38, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 4707.4294087917015}

Rank 6: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.027142230268205066, 'reg__l2_leaf_reg': 22.68575762902154, 'reg__bagging_temperature': 3.132704138863625, 'reg__random_strength': 2.307099023594934, 'reg__subsample': 0.7850379976334958, 'reg__min_data_in_leaf': 20, 'reg__leaf_estimation_iterations': 6, 'reg__rsm': 0.6169788585833644, 'reg__border_count': 34, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 2755.053859689628}

Rank 7: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.03446193426350863, 'reg__l2_leaf_reg': 28.14699974372354, 'reg__bagging_temperature': 1.4268005239705999, 'reg__random_strength': 2.7924189491030935, 'reg__subsample': 0.8244434296305433, 'reg__min_data_in_leaf': 10, 'reg__leaf_estimation_iterations': 5, 'reg__rsm': 0.701564250985659, 'reg__border_count': 51, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 5304.20646719794}

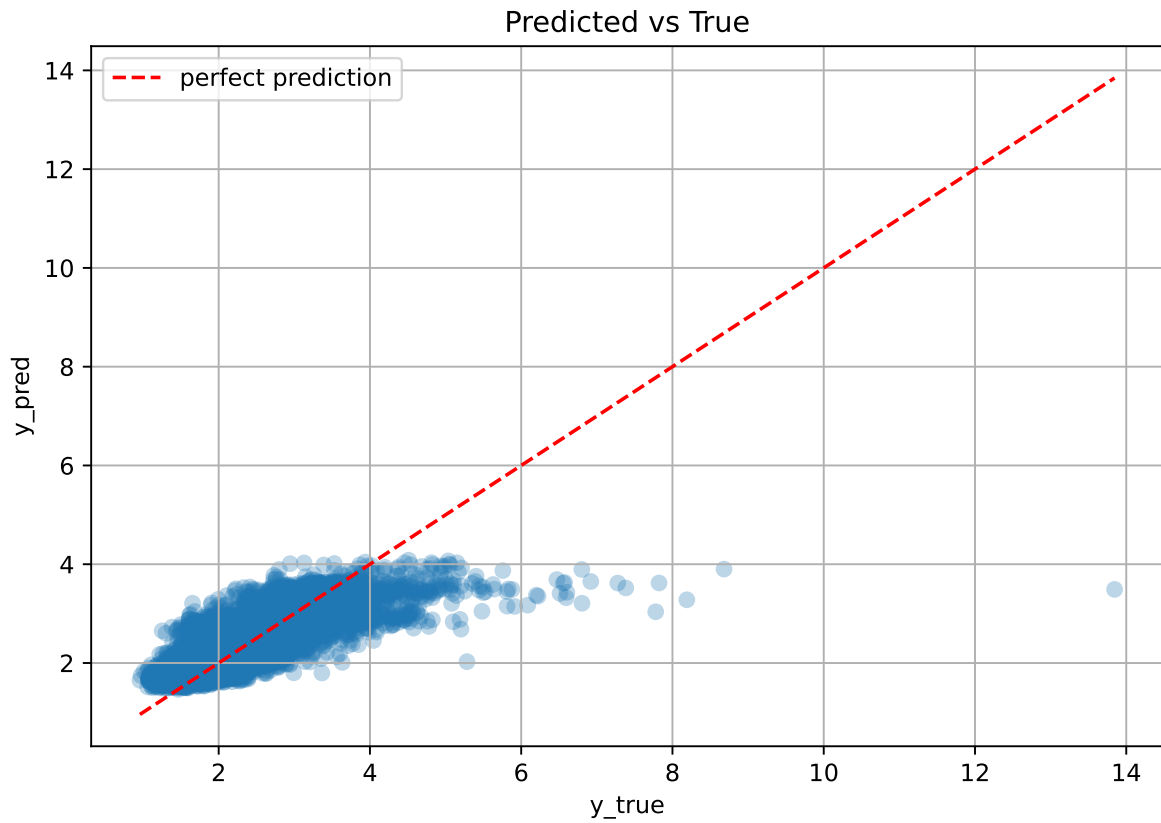
Rank 8: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.022146895418563534, 'reg__l2_leaf_reg': 18.98923512603672, 'reg__bagging_temperature': 3.5399580393980474, 'reg__random_strength': 2.753117082996621, 'reg__subsample': 0.839627612368725, 'reg__min_data_in_leaf': 24, 'reg__leaf_estimation_iterations': 8, 'reg__rsm': 0.5742299235459096, 'reg__border_count': 46, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 2433.662058320087}

Rank 9: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.021510242165990238, 'reg__l2_leaf_reg': 7.481908131895451, 'reg__bagging_temperature': 3.5861185888014124, 'reg__random_strength': 2.7274729488267133, 'reg__subsample': 0.7685362422906268, 'reg__min_data_in_leaf': 24, 'reg__leaf_estimation_iterations': 8, 'reg__rsm': 0.5710426261893815, 'reg__border_count': 47, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 1561.2225556618127}

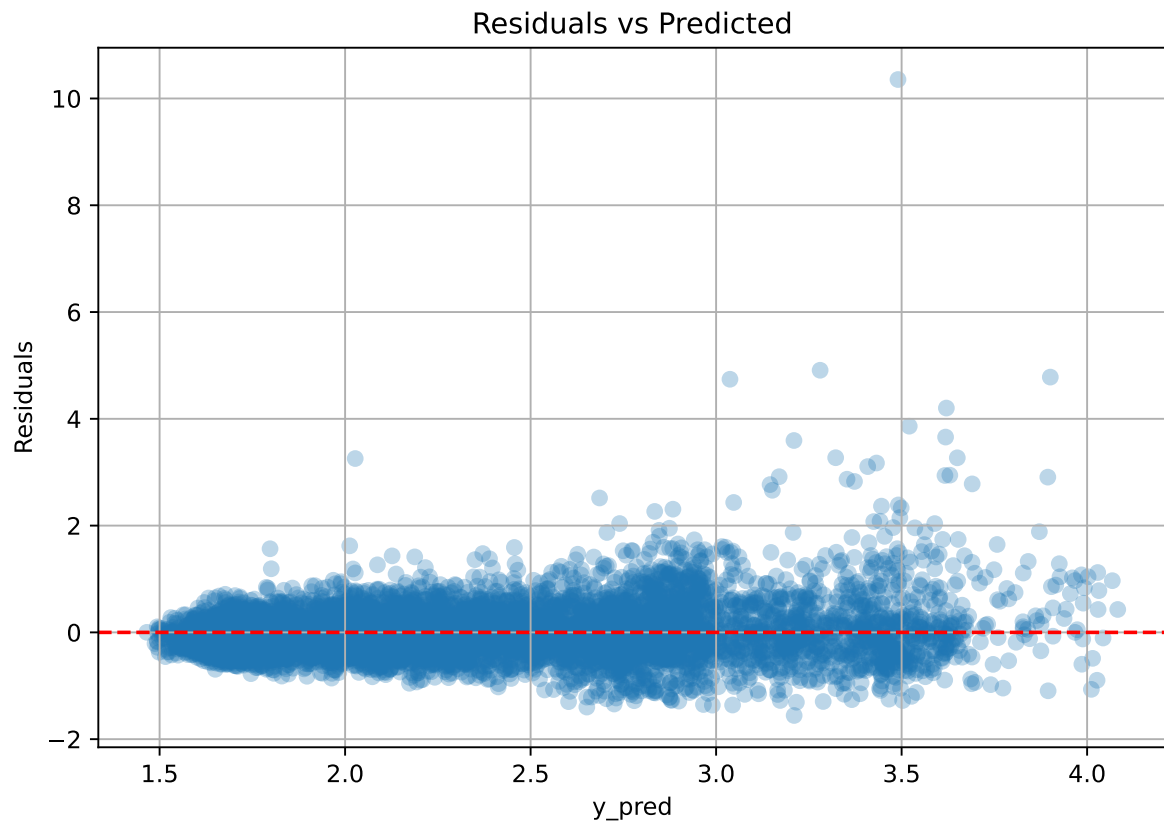
Rank 10: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.029025899071209144, 'reg__l2_leaf_reg': 20.959035898066595, 'reg__bagging_temperature': 3.018910363305157,

```
'reg__random_strength': 2.3163051205248704, 'reg__subsample': 0.7849977858253967,  
'reg__min_data_in_leaf': 20, 'reg__leaf_estimation_iterations': 6, 'reg__rsm':  
0.6240526957385328, 'reg__border_count': 33, 'reg__leaf_estimation_method': 'Gradient',  
'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_tempera-  
ture': 4743.77124378232}
```

3.6.2 Scatter



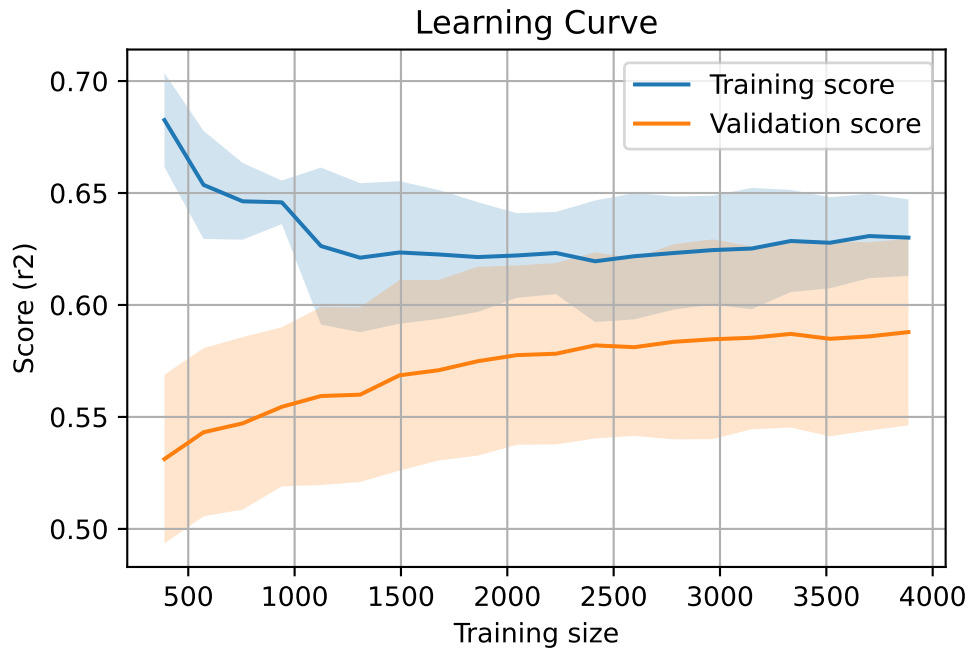
3.6.3 Scatter residuals



3.6.4 Learning curve

`Len(group)` : 7932, `Len(training)` : 5553,

`Len(training_real)` : 3888, `Len(test_of_training)` : 1665, `Len(test)` : 2379



3.7 Sumup-table

model	mean_train_cv	mean_test_cv	std_test_cv	test_cv	test	diff	r_test	r_train	pen_score
Linear	0.522	0.456	0.074	0.537	0.537	0.081	1.177	1.143	nan
LGBM	0.624	0.579	0.042	0.614	0.636	0.057	1.098	1.079	0.3501
CatBoost	0.631	0.59	0.042	0.63	0.636	0.046	1.078	1.07	0.3846

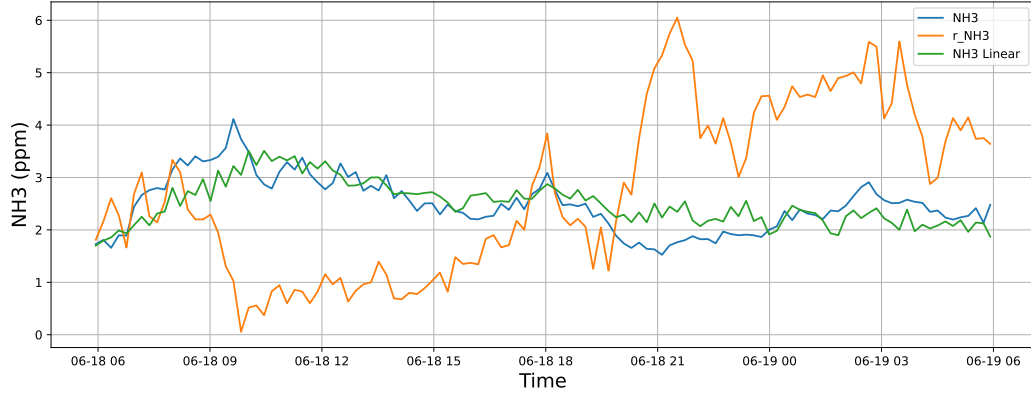
4 Plot

4.1 Standalone plots

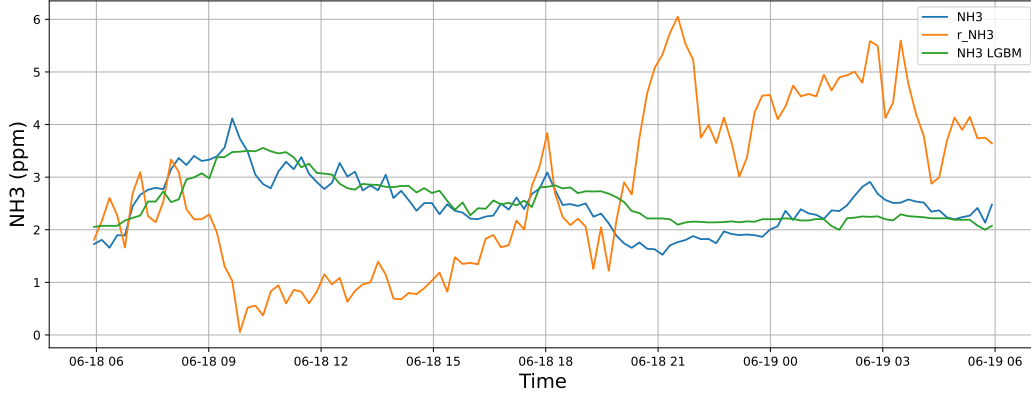
4.1.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

4.1.1.1 Time window : 24

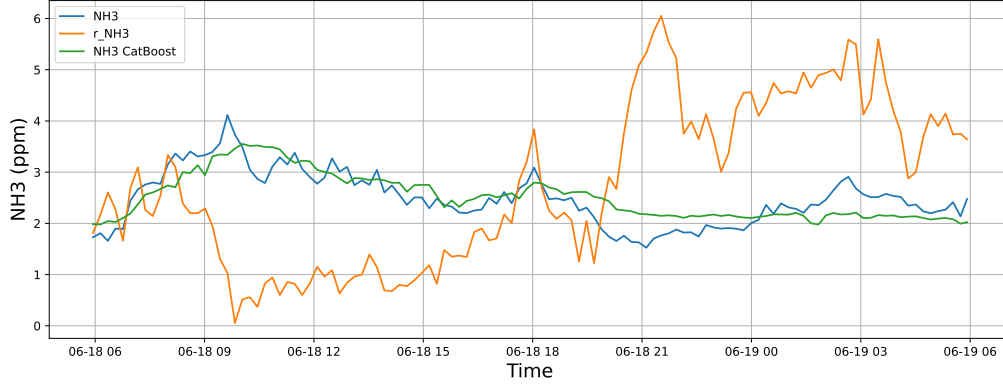
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

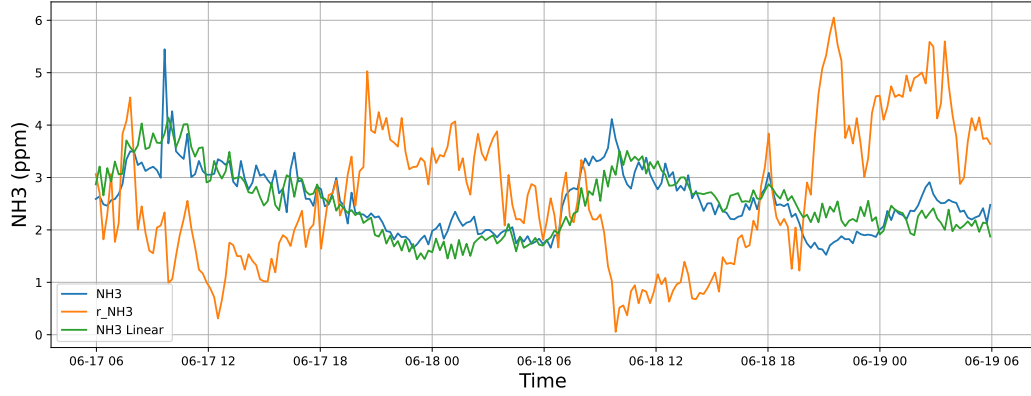


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

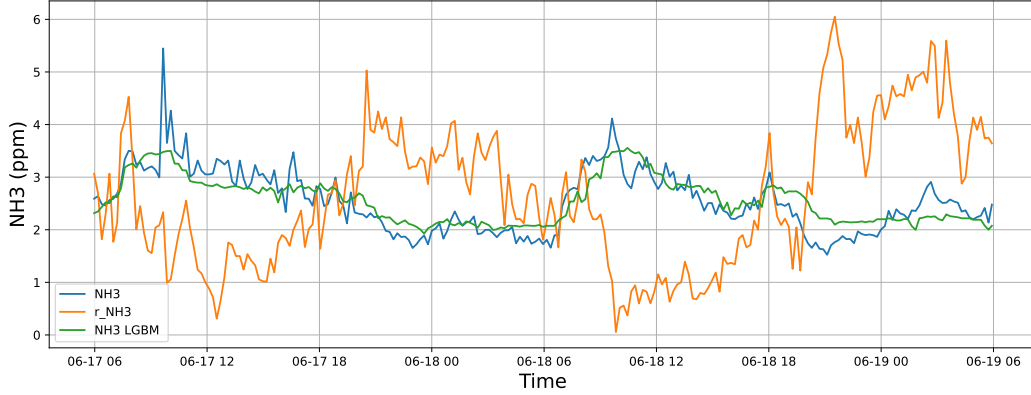


4.1.1.2 Time window : 48

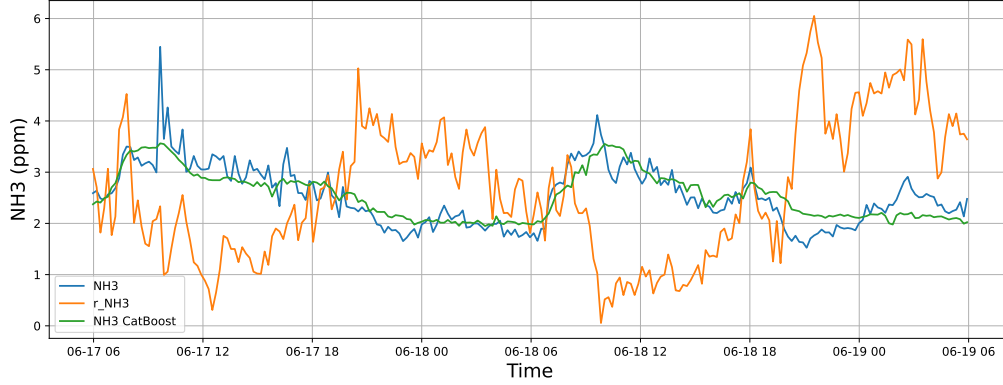
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

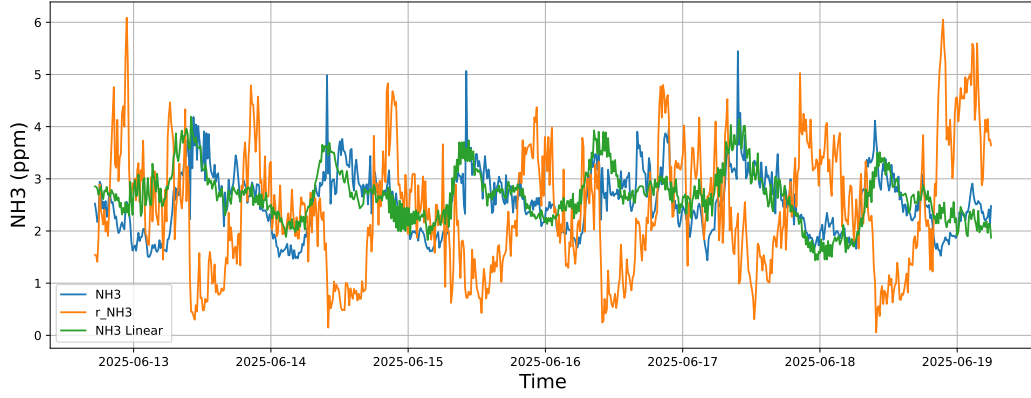


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

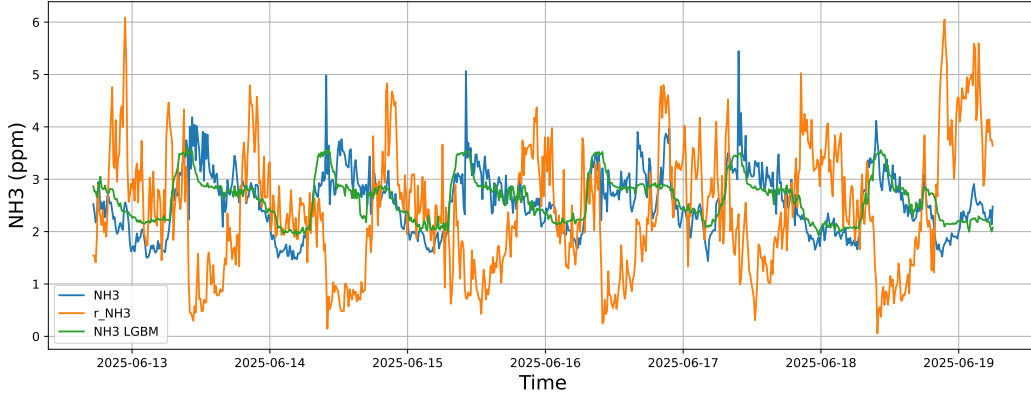


4.1.1.3 Time window : ALL

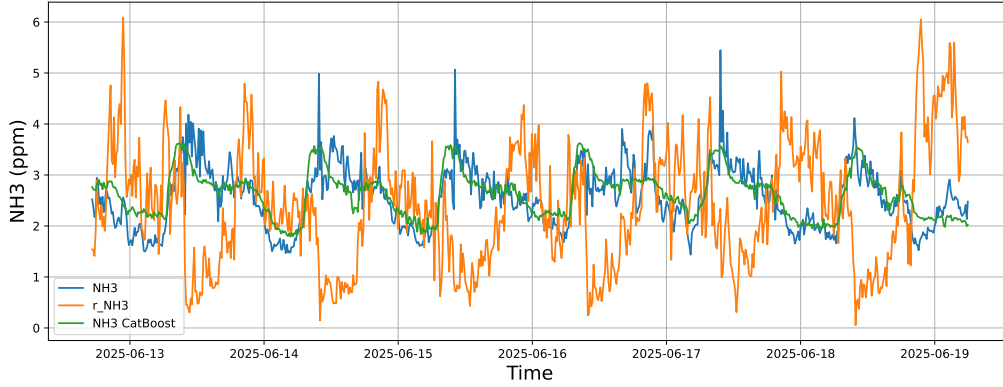
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



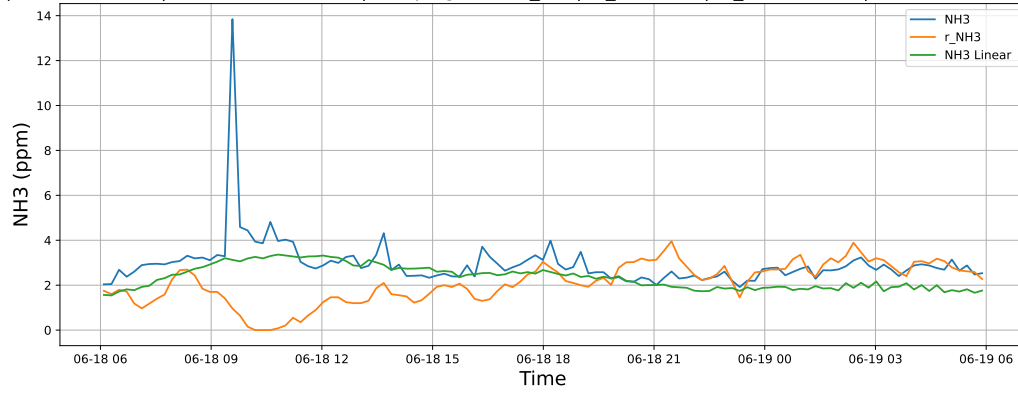
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



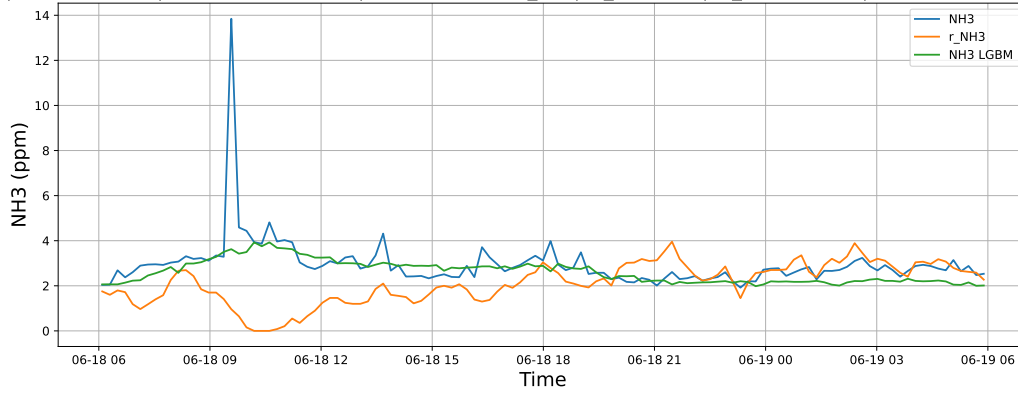
4.1.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

4.1.2.1 Time window : 24

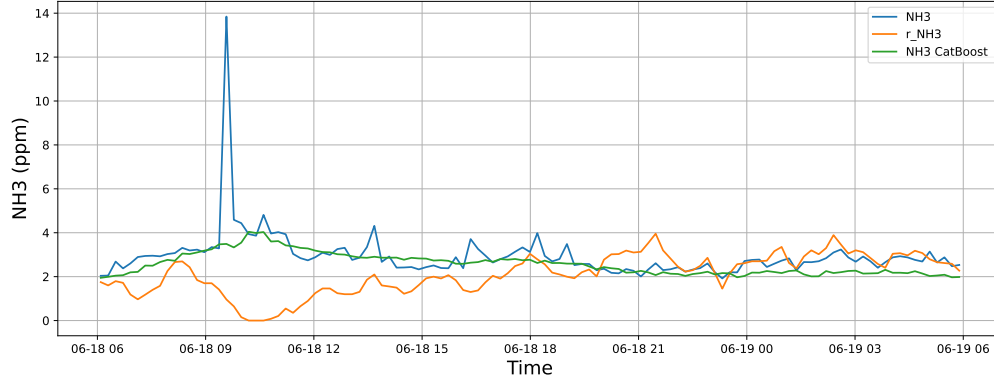
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

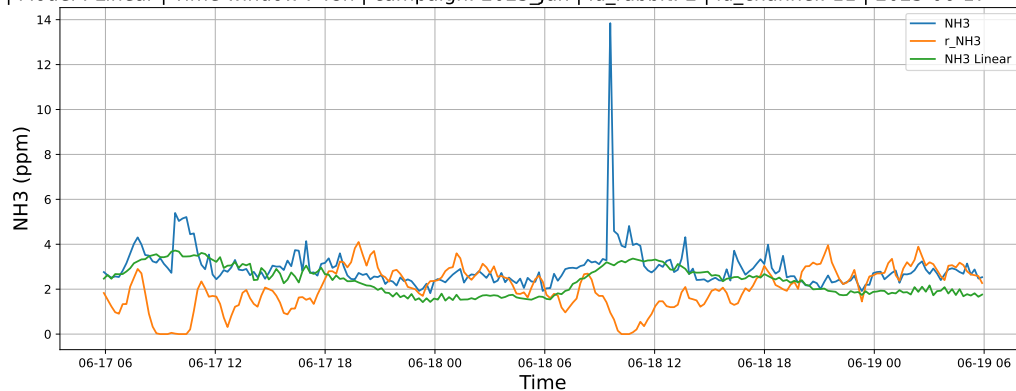


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

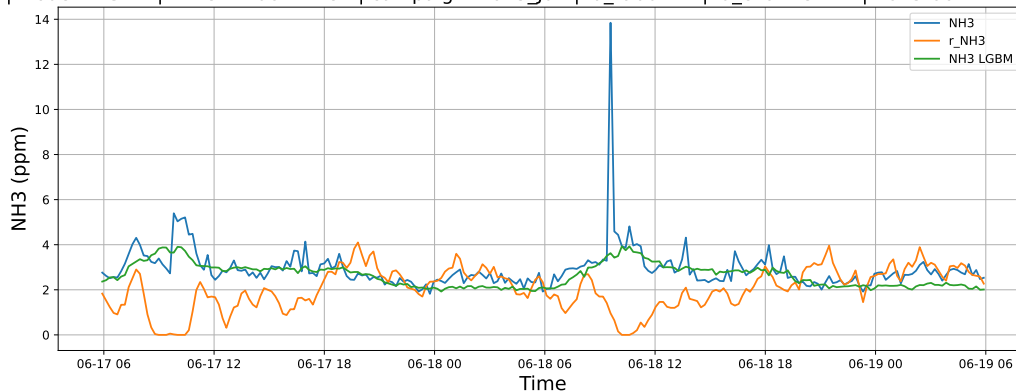


4.1.2.2 Time window : 48

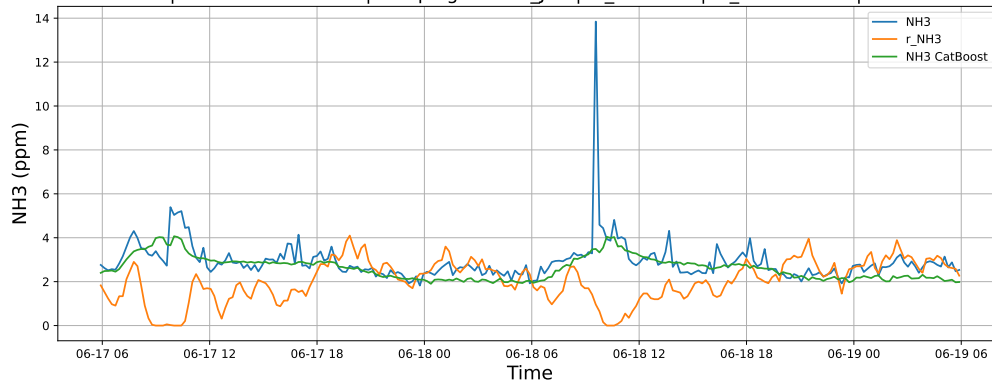
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

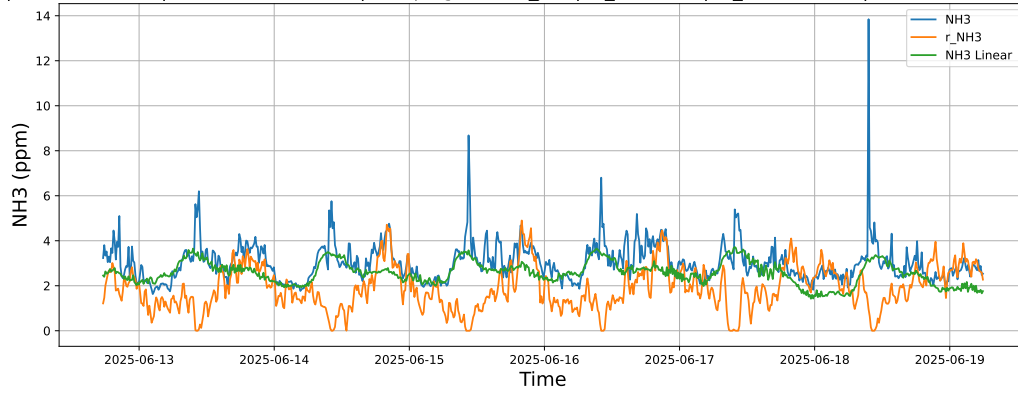


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

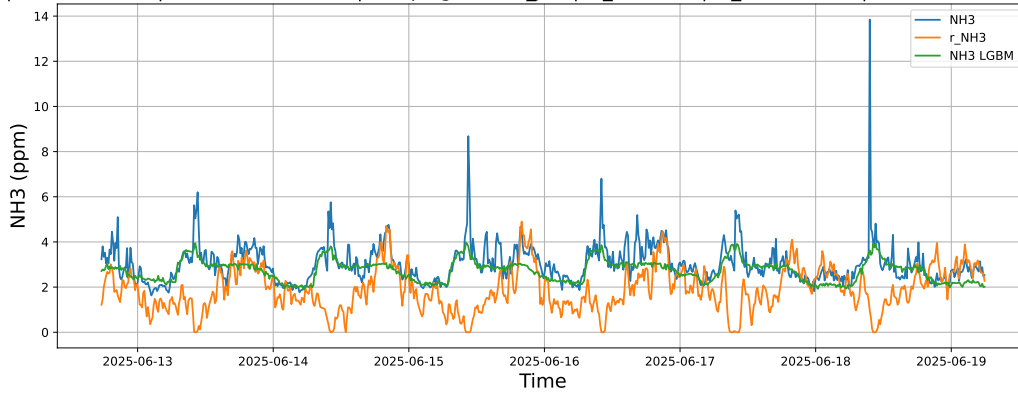


4.1.2.3 Time window : ALL

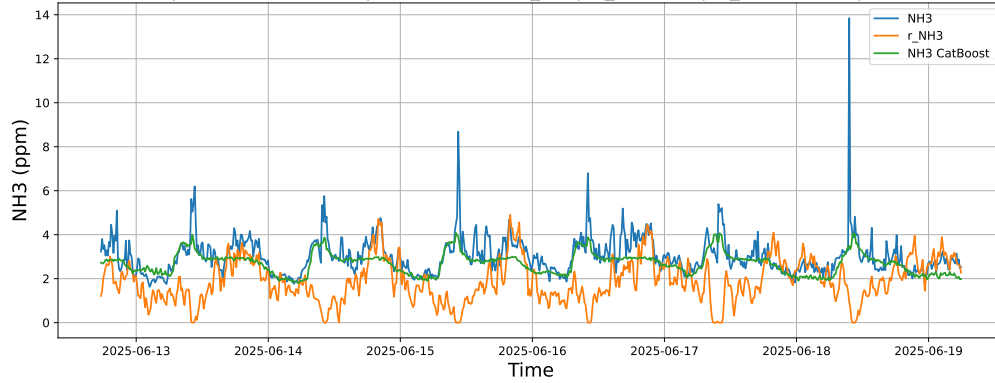
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



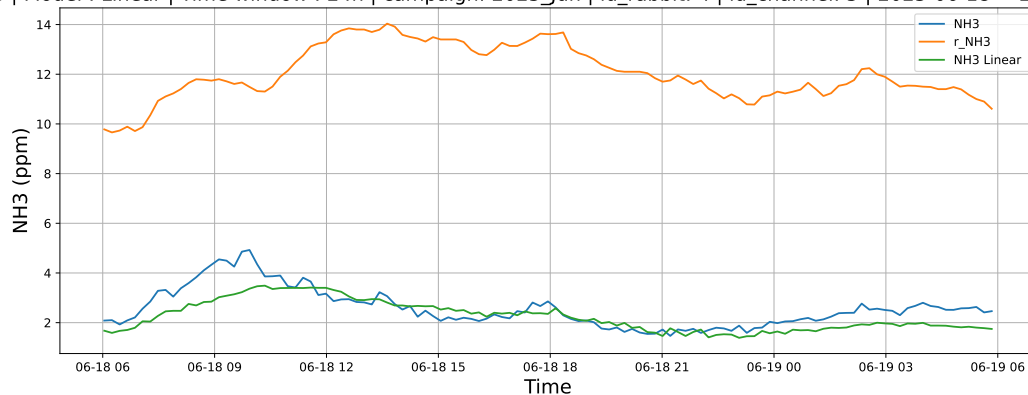
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



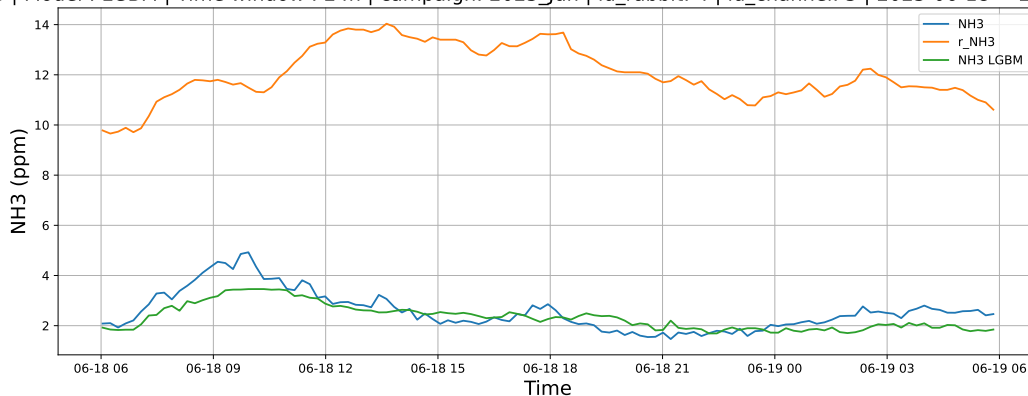
4.1.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

4.1.3.1 Time window : 24

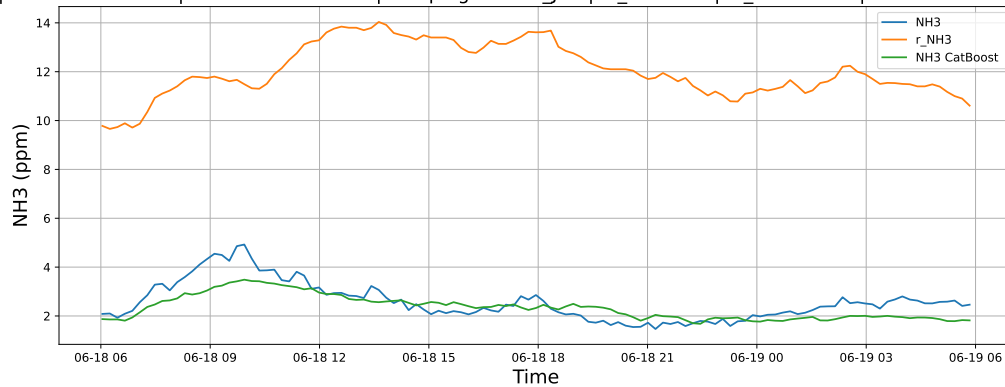
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

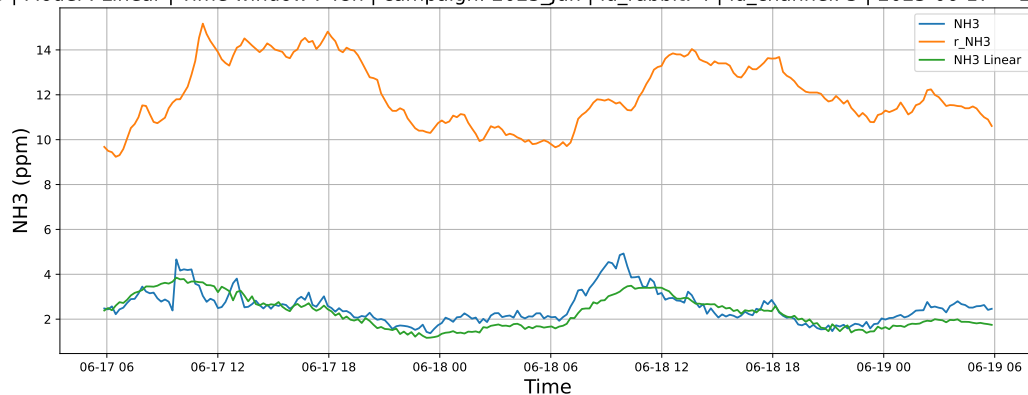


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

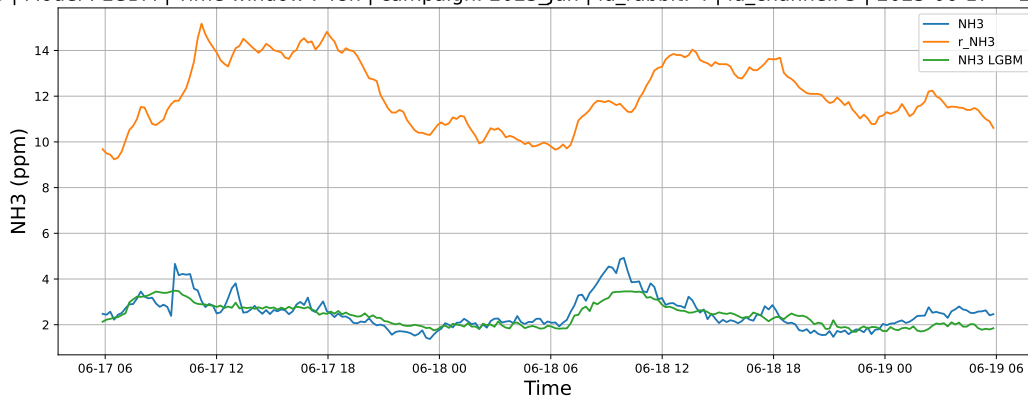


4.1.3.2 Time window : 48

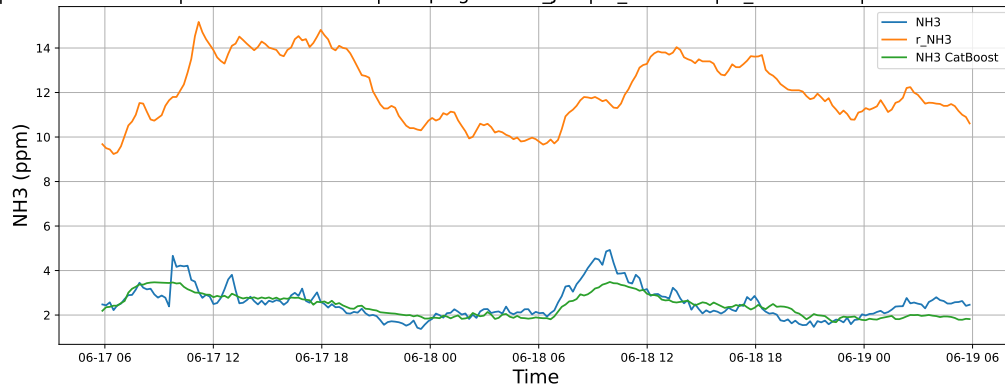
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

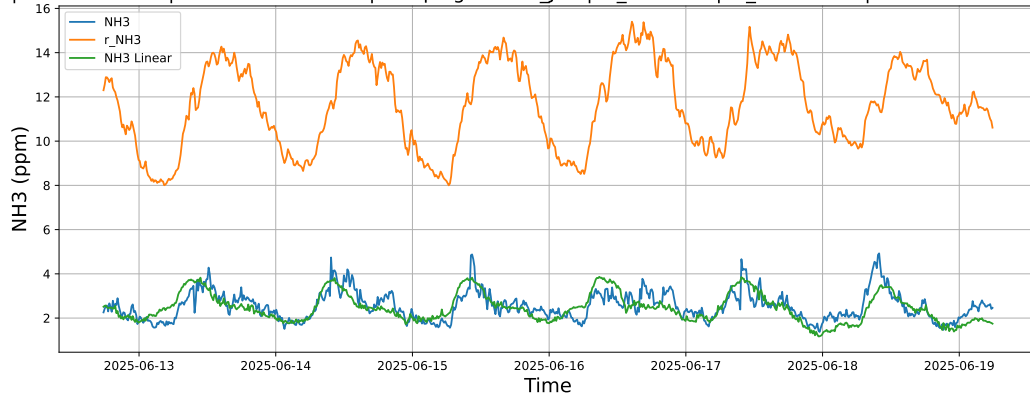


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

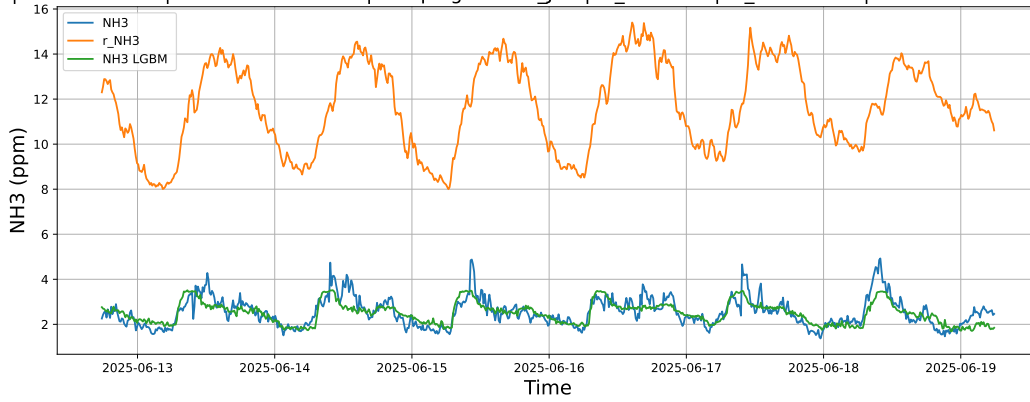


4.1.3.3 Time window : ALL

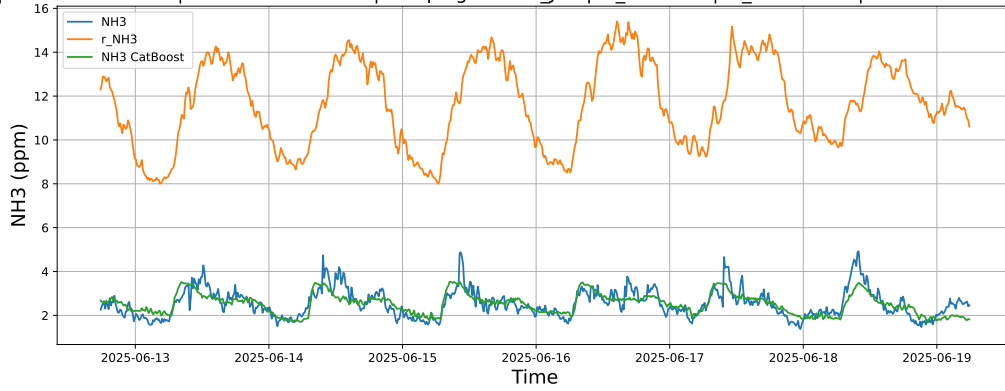
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



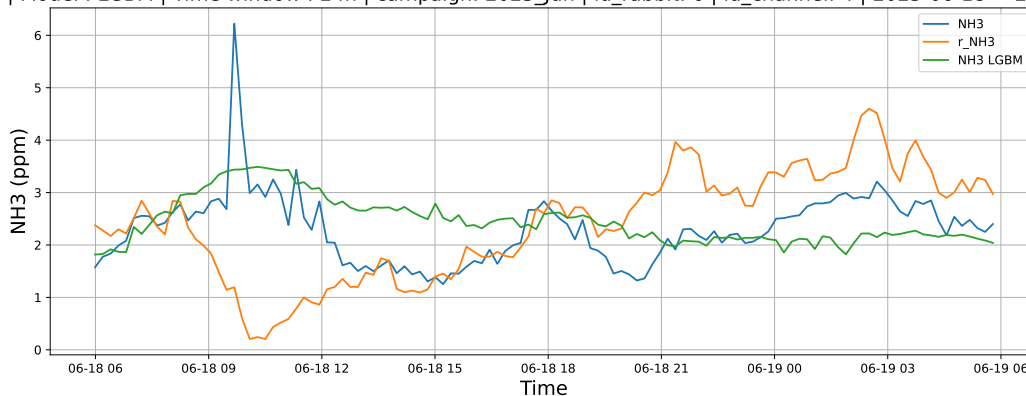
4.1.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

4.1.4.1 Time window : 24

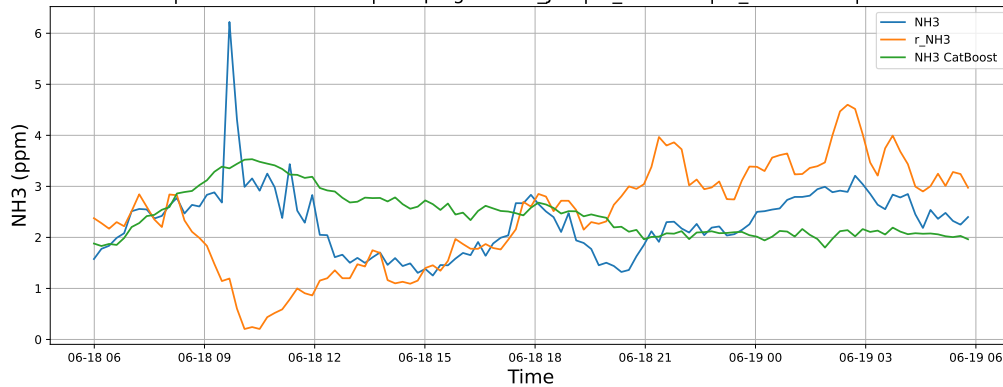
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

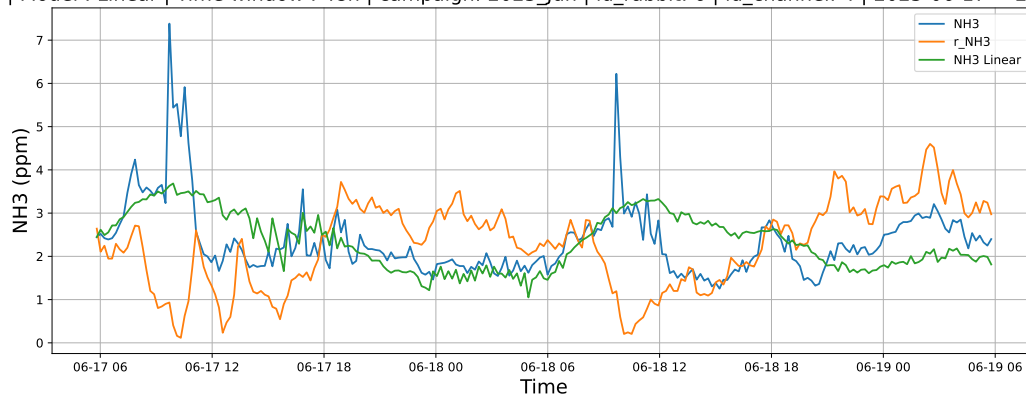


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

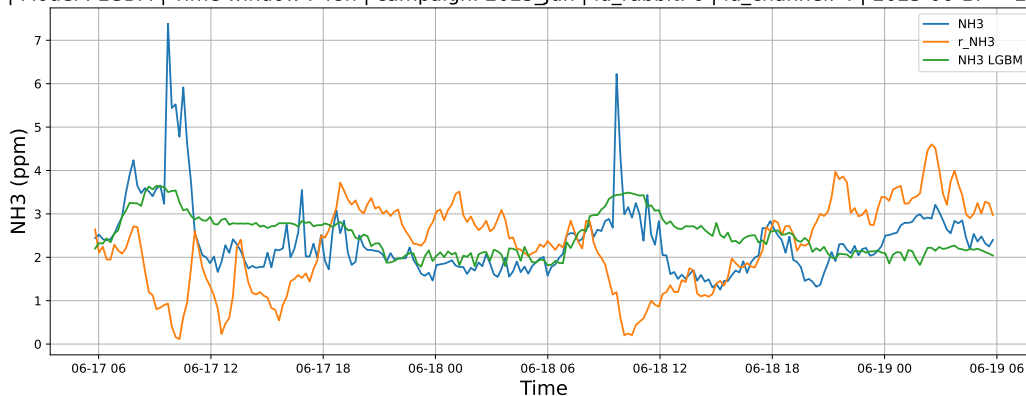


4.1.4.2 Time window : 48

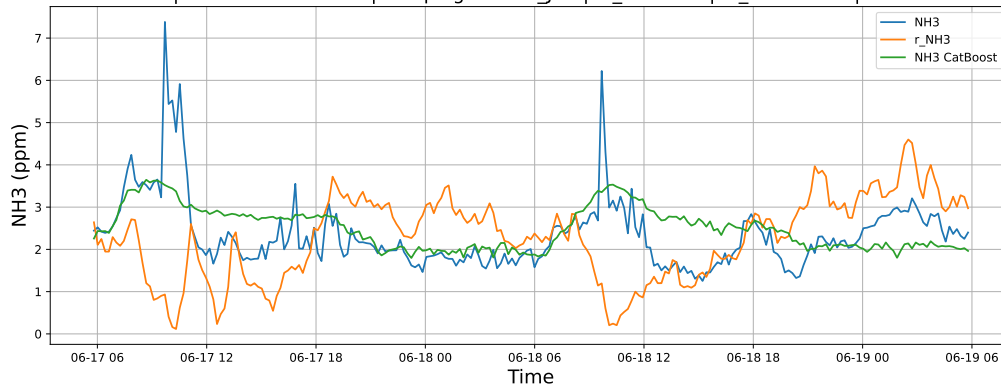
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

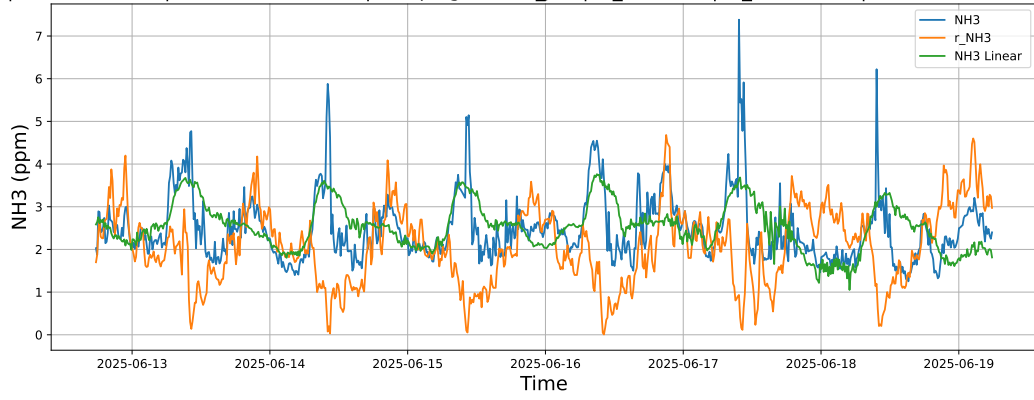


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

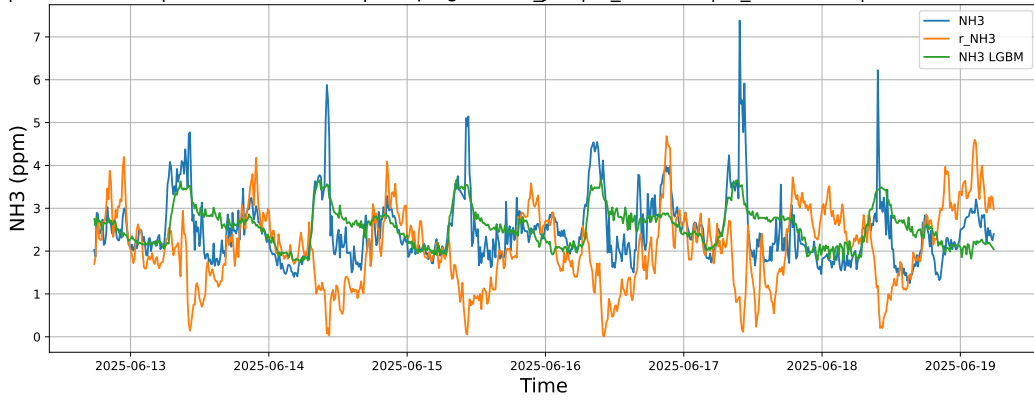


4.1.4.3 Time window : ALL

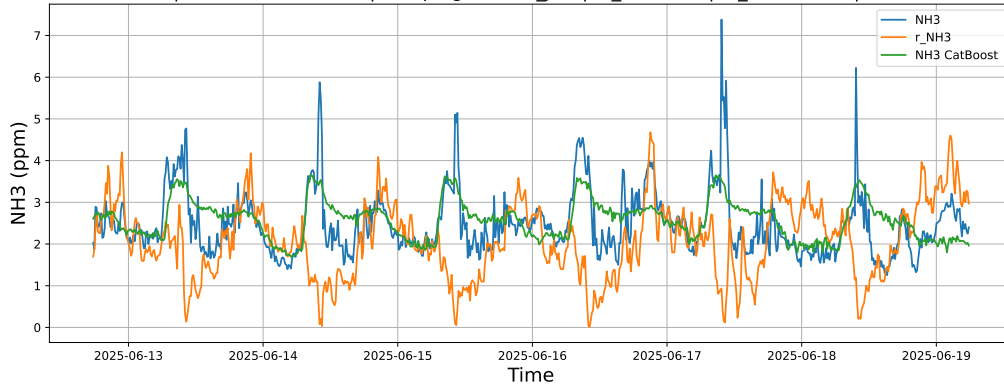
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



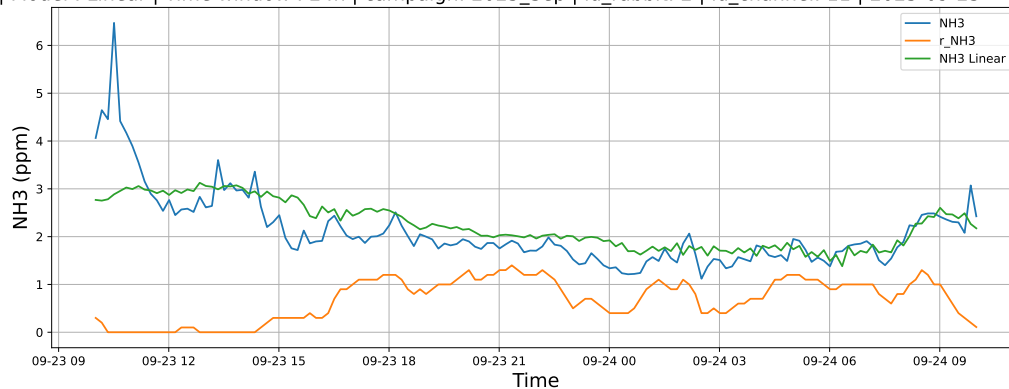
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



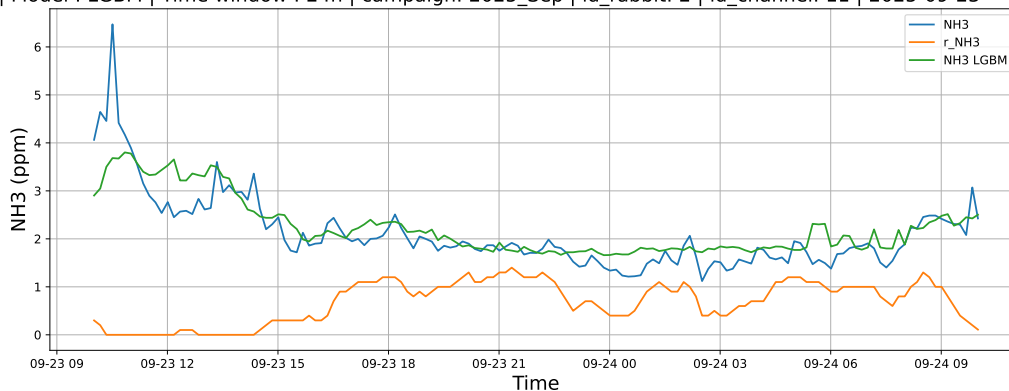
4.1.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

4.1.5.1 Time window : 24

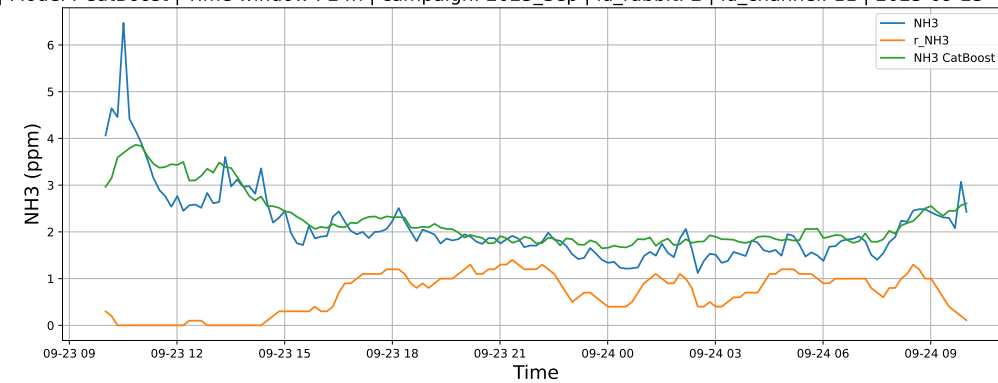
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

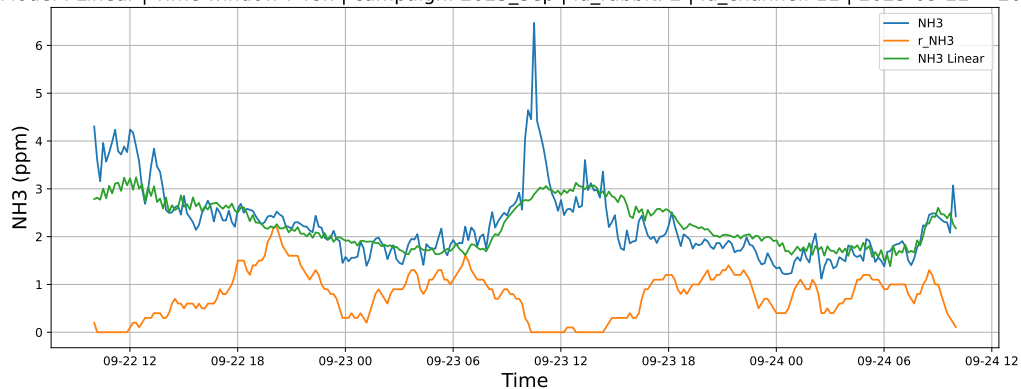


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

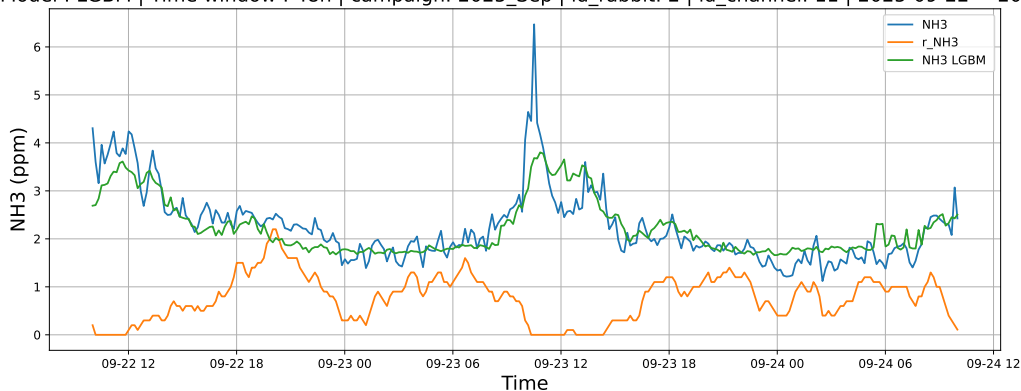


4.1.5.2 Time window : 48

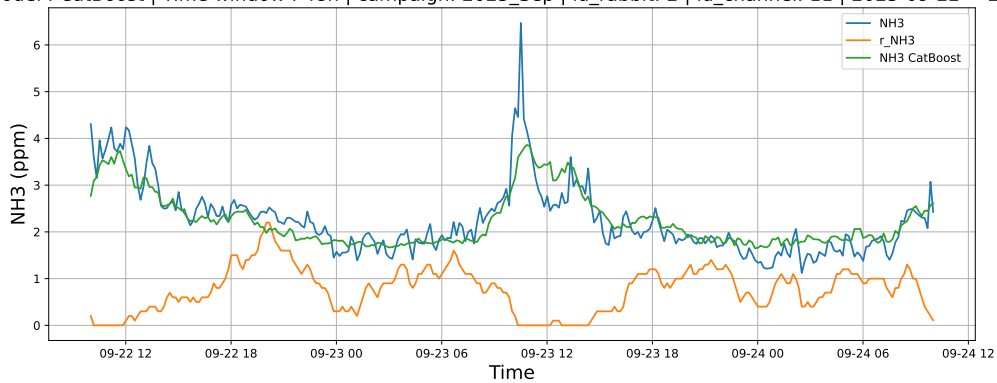
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

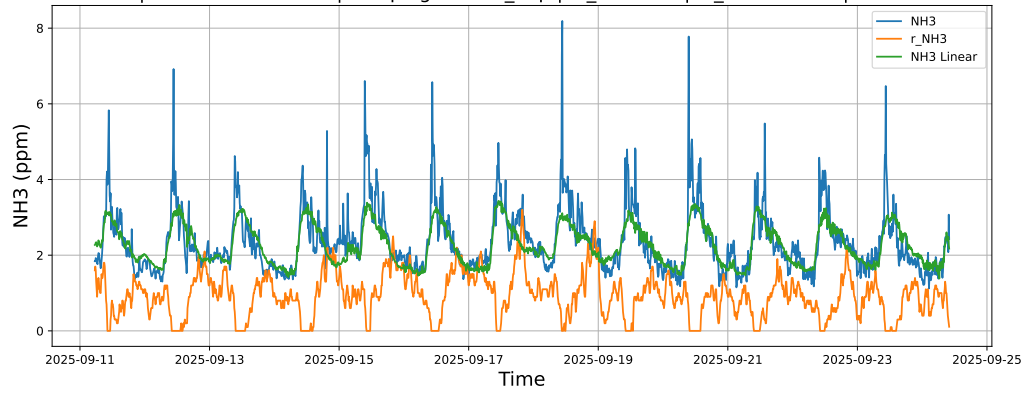


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

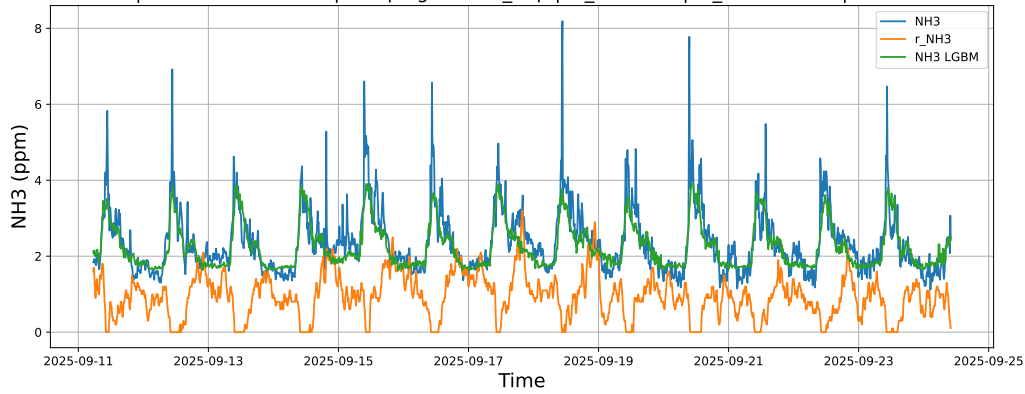


4.1.5.3 Time window : ALL

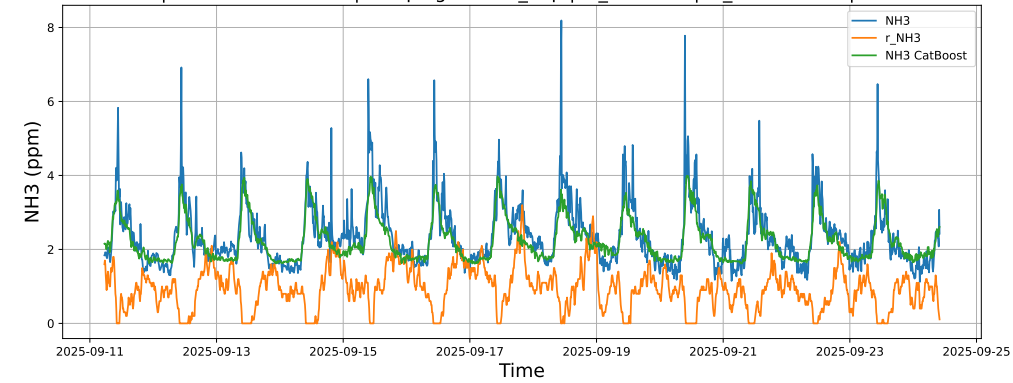
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



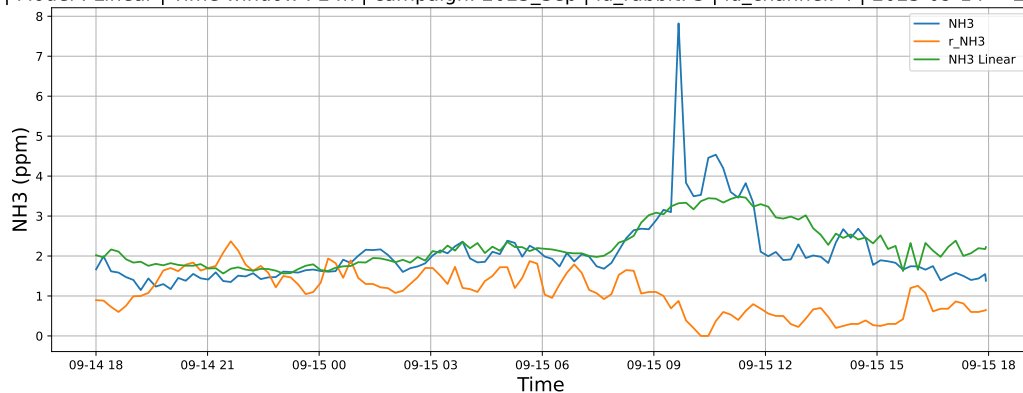
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



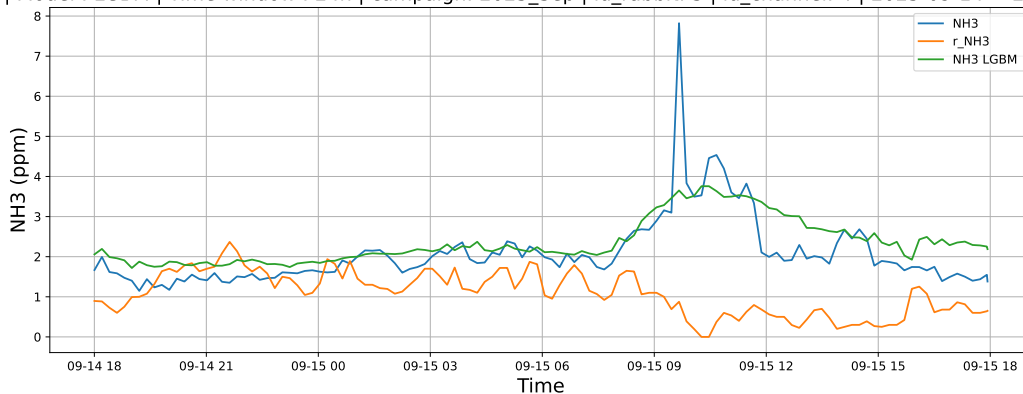
4.1.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

4.1.6.1 Time window : 24

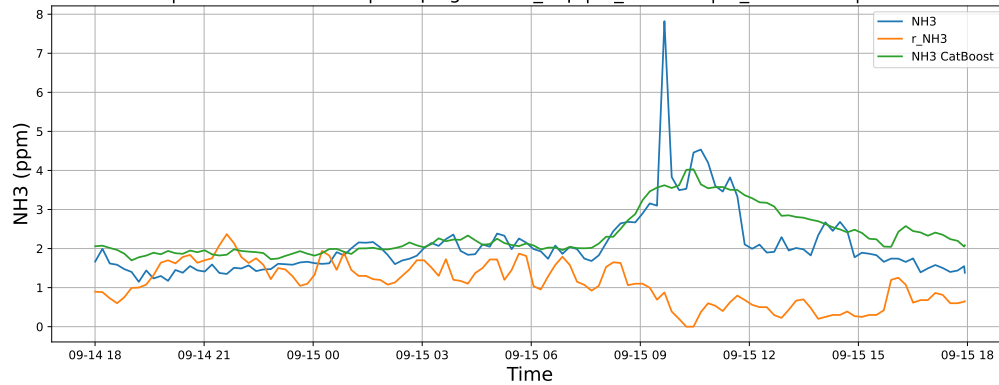
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

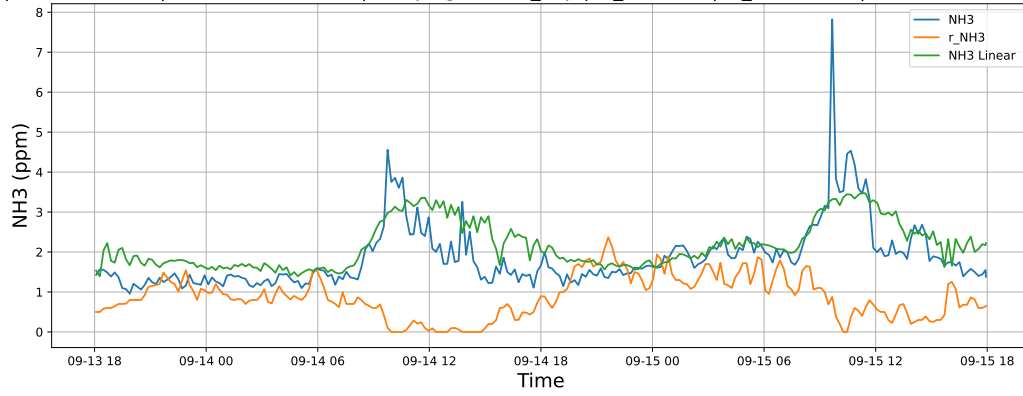


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

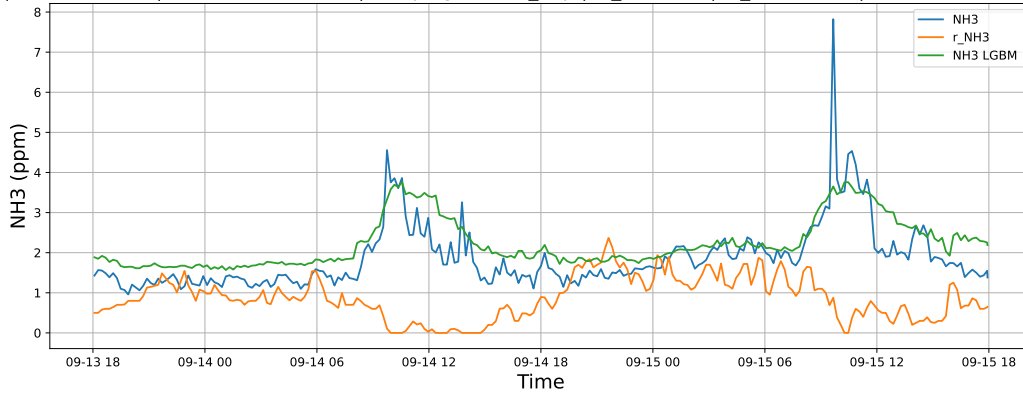


4.1.6.2 Time window : 48

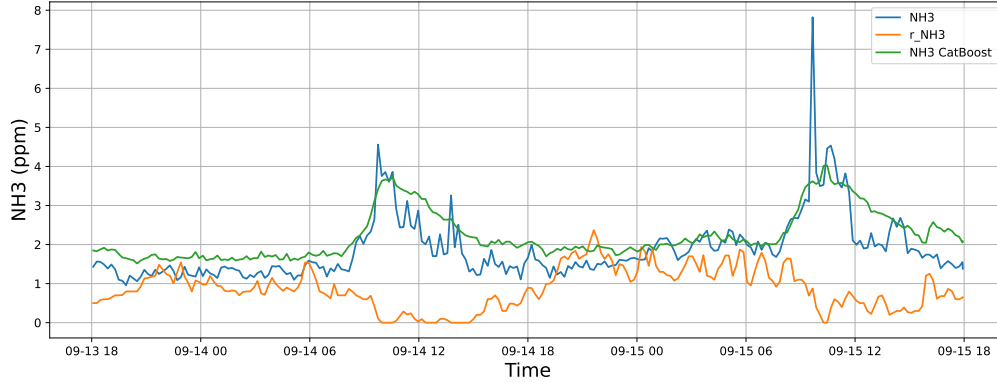
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

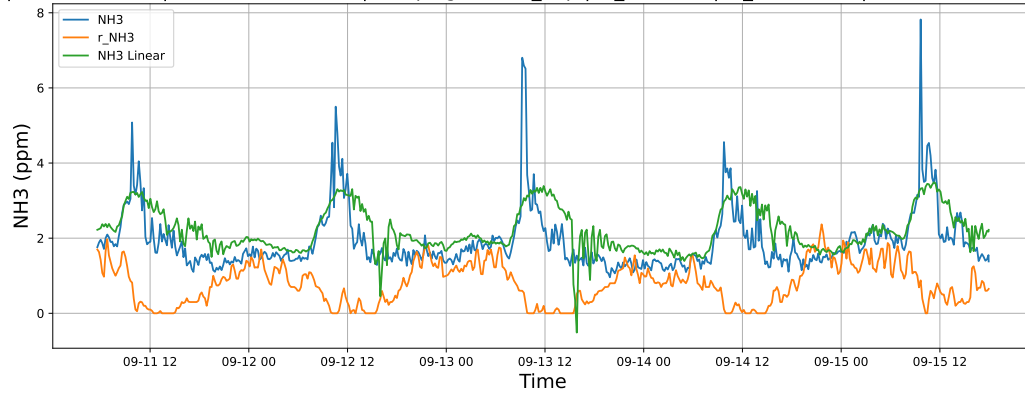


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

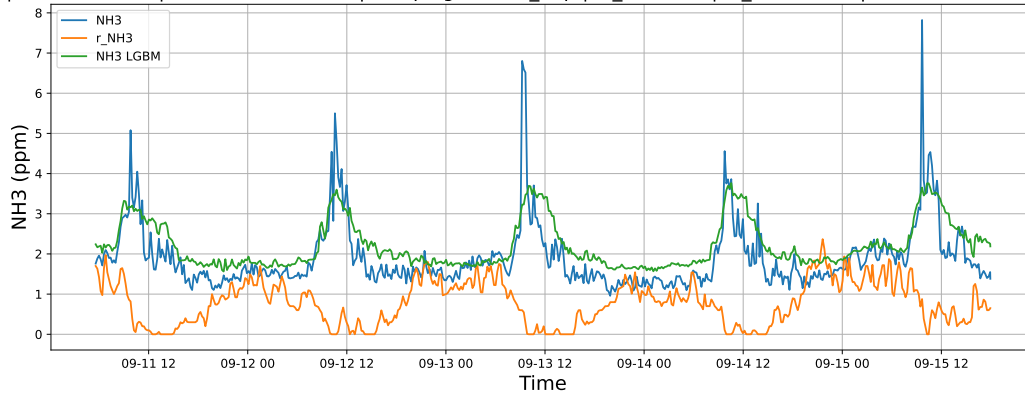


4.1.6.3 Time window : ALL

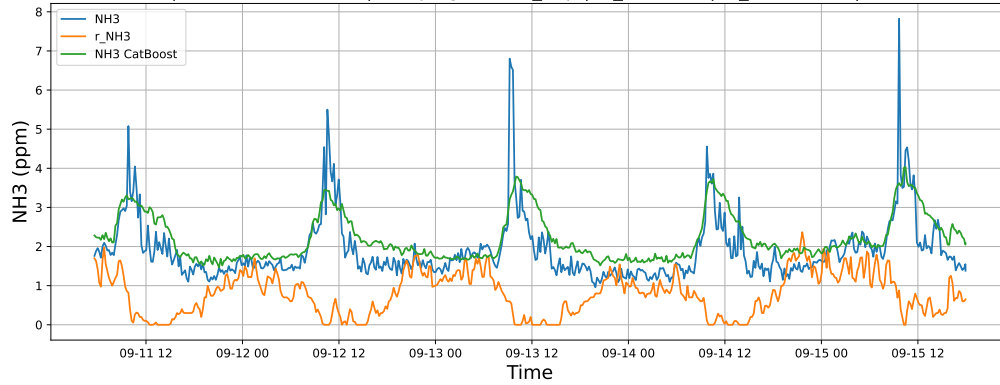
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



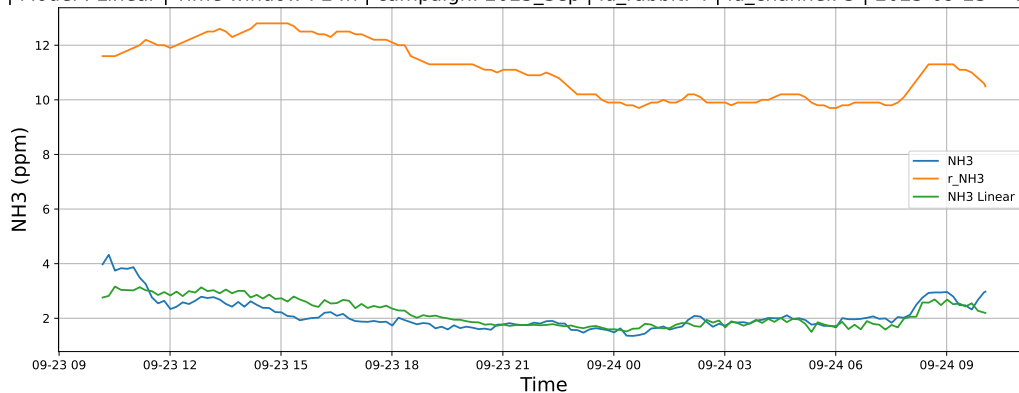
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



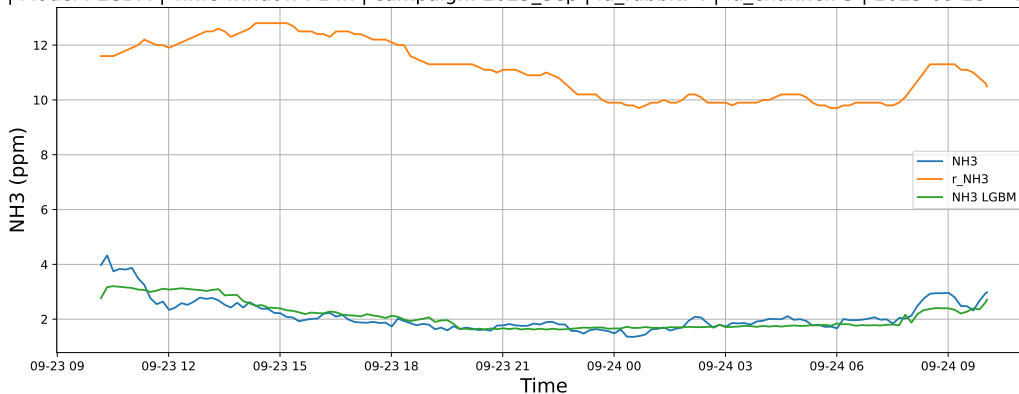
4.1.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

4.1.7.1 Time window : 24

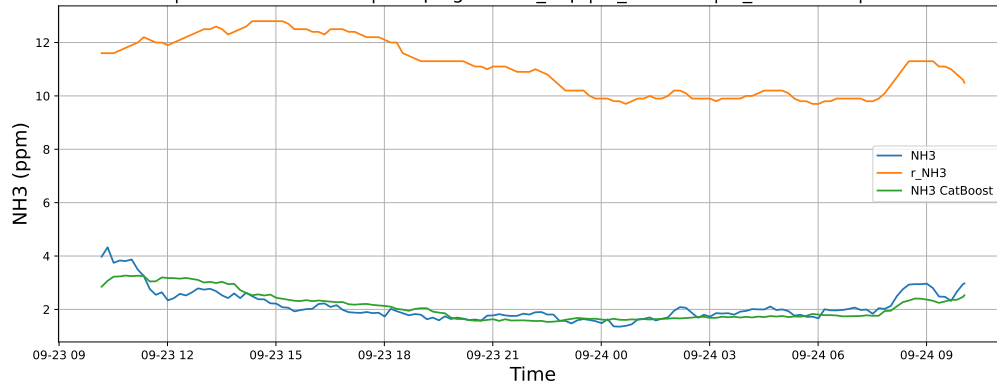
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

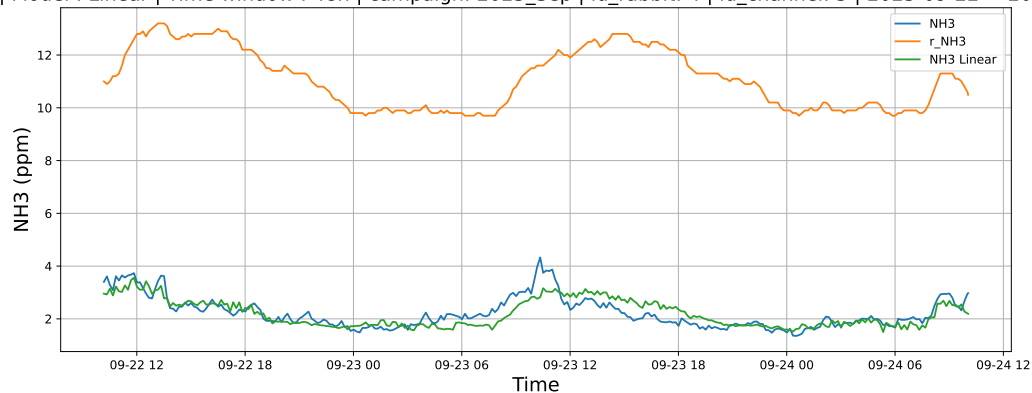


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

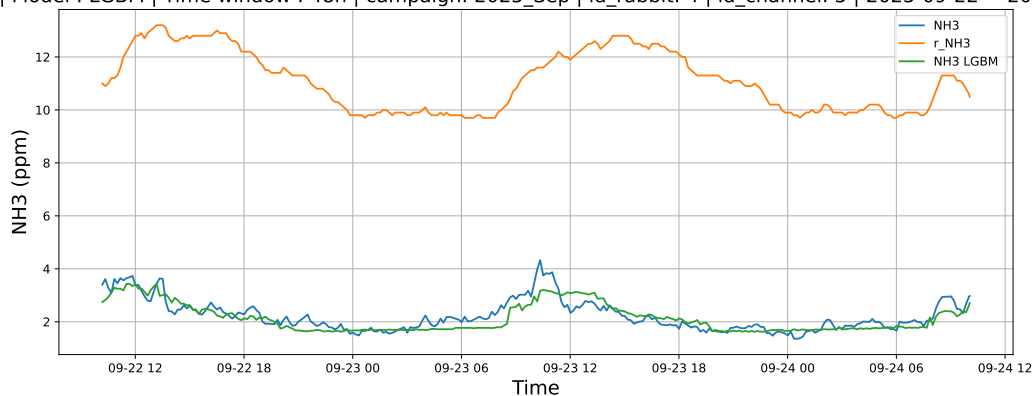


4.1.7.2 Time window : 48

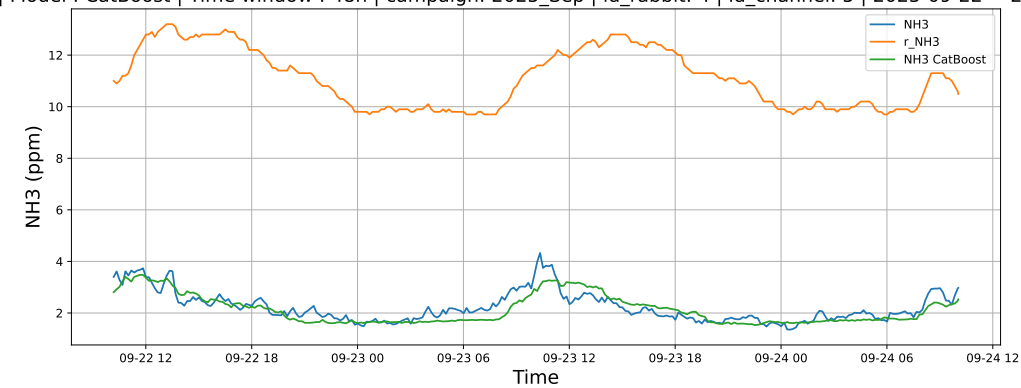
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

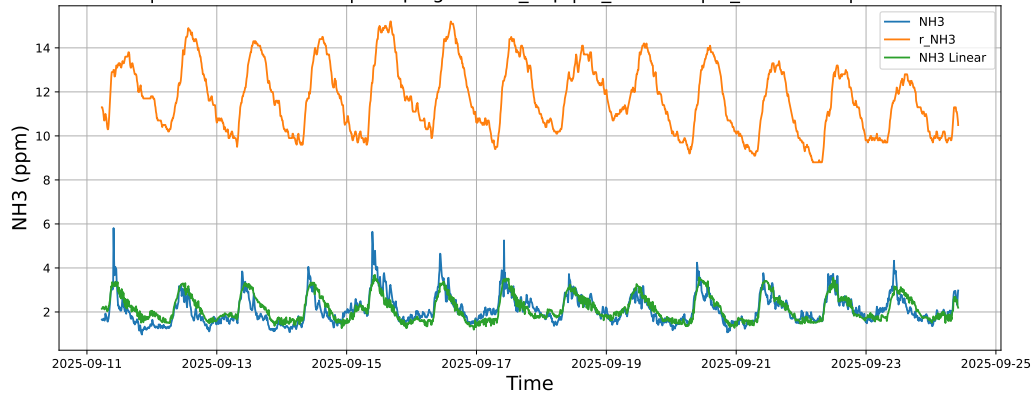


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

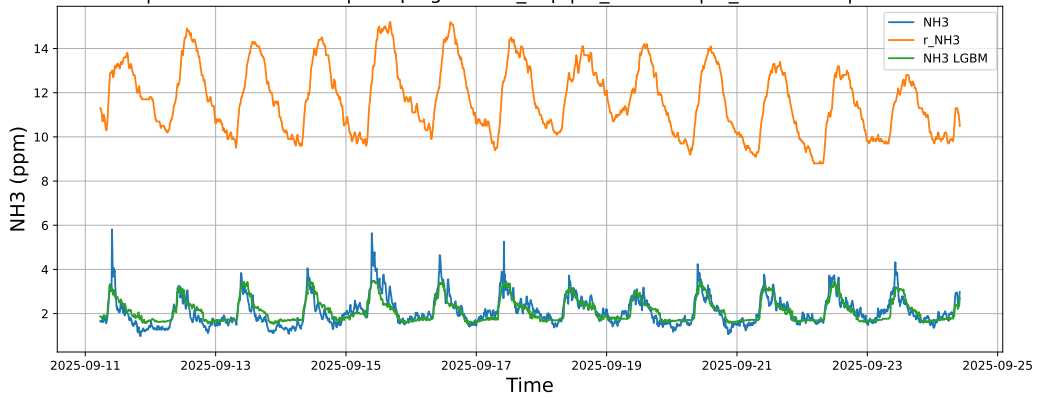


4.1.7.3 Time window : ALL

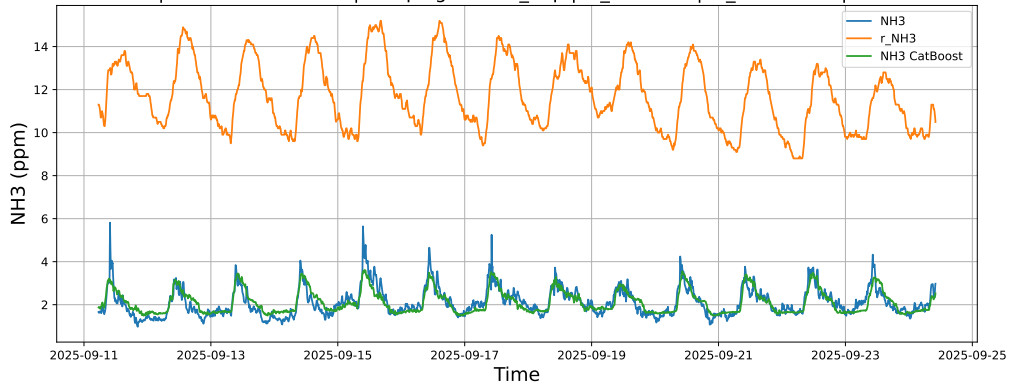
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



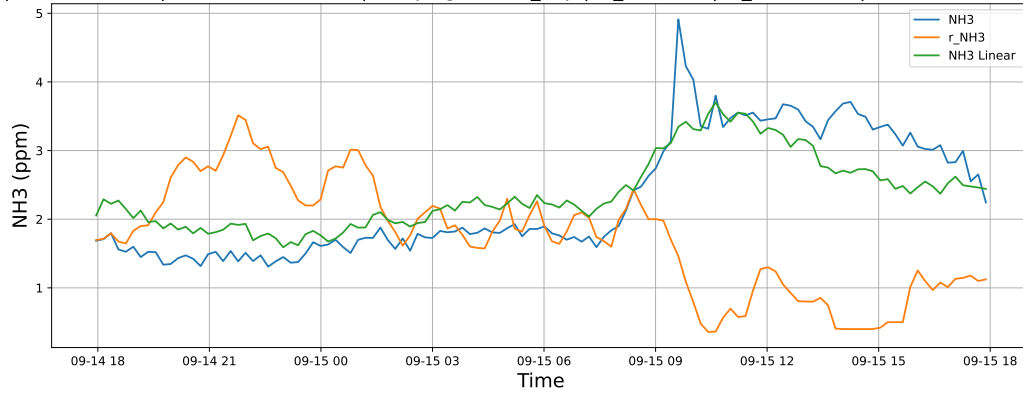
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



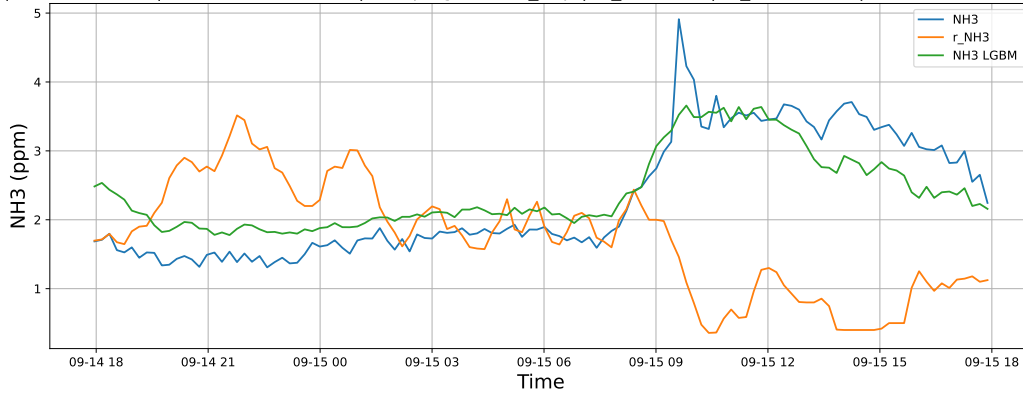
4.1.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

4.1.8.1 Time window : 24

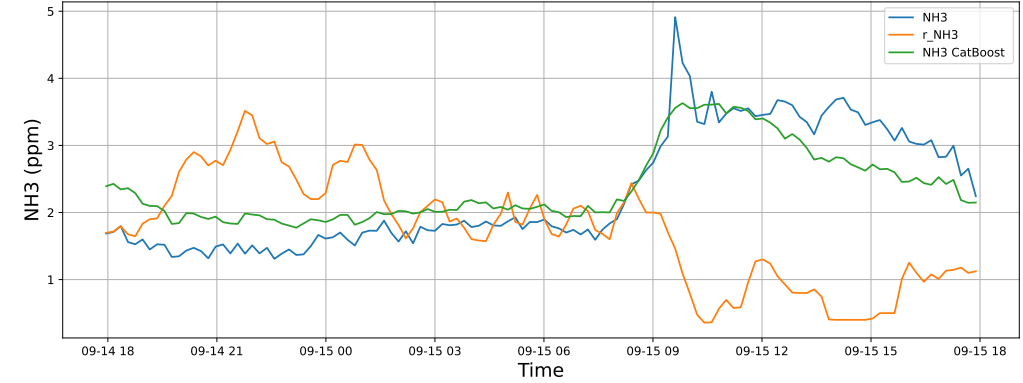
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

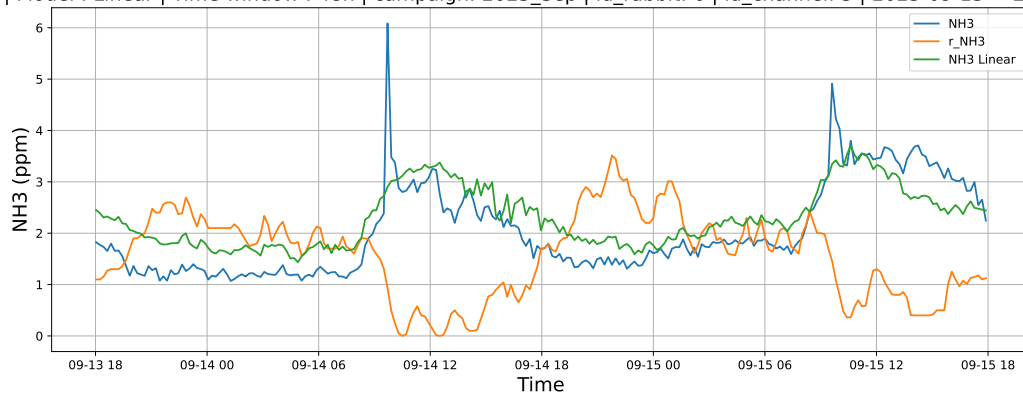


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

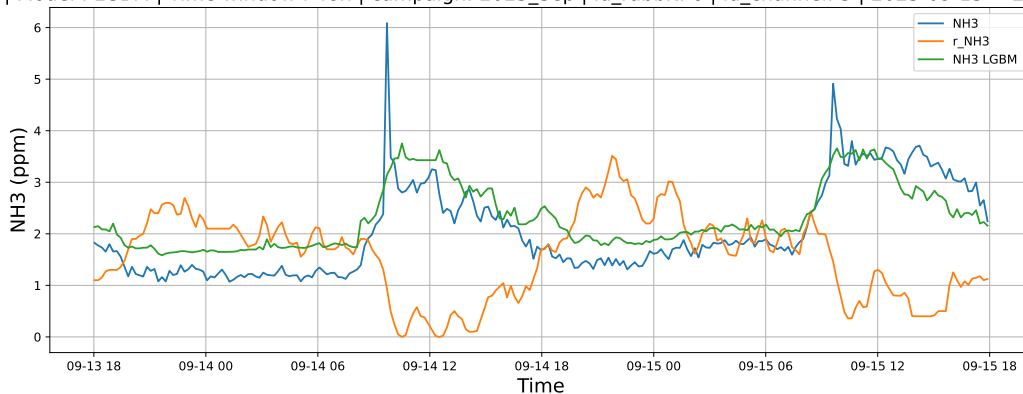


4.1.8.2 Time window : 48

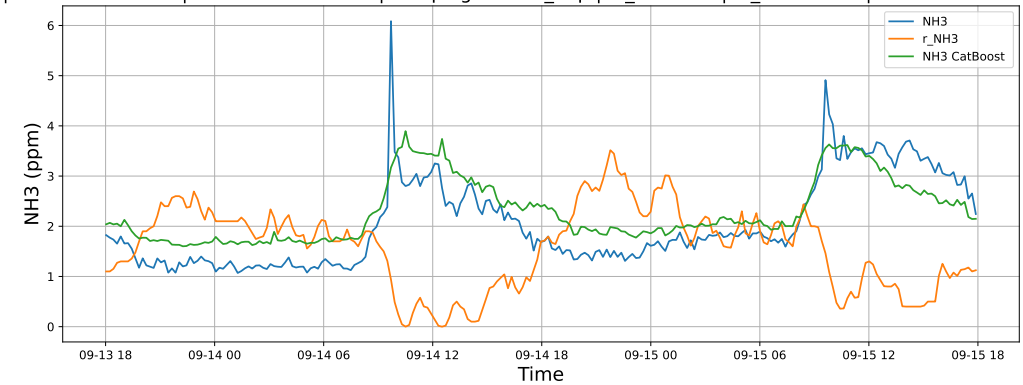
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

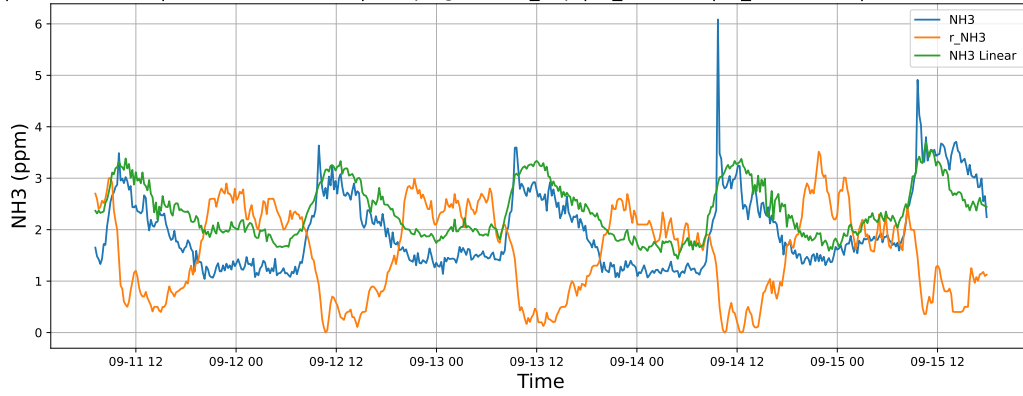


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

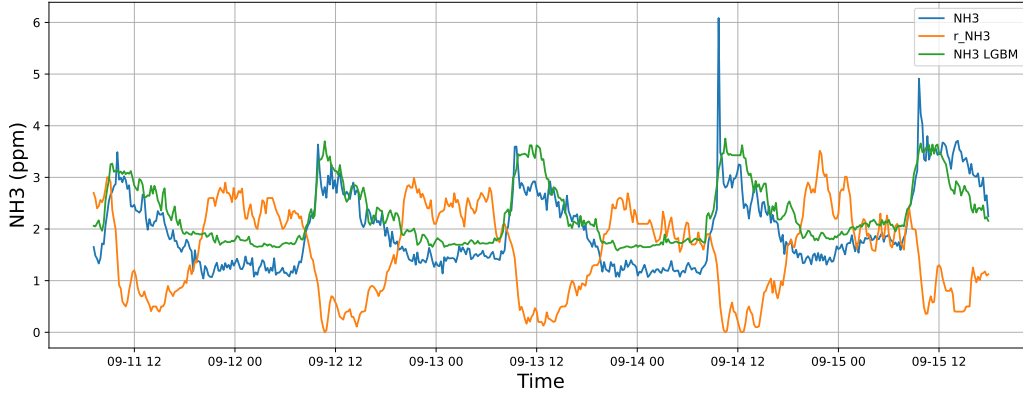


4.1.8.3 Time window : ALL

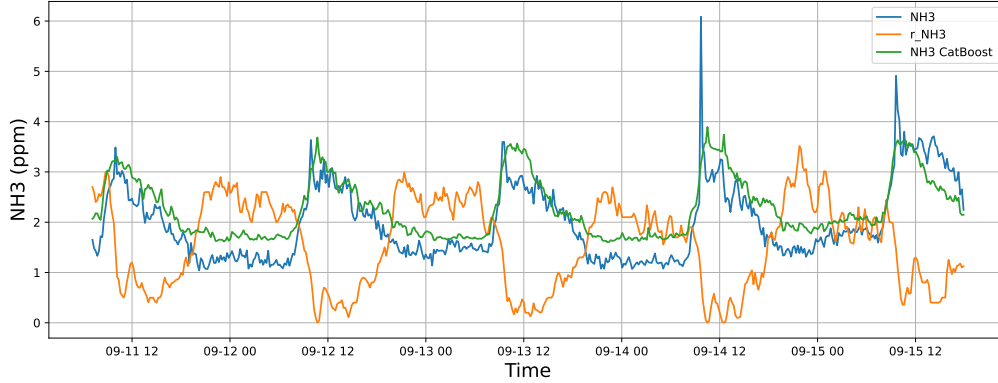
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

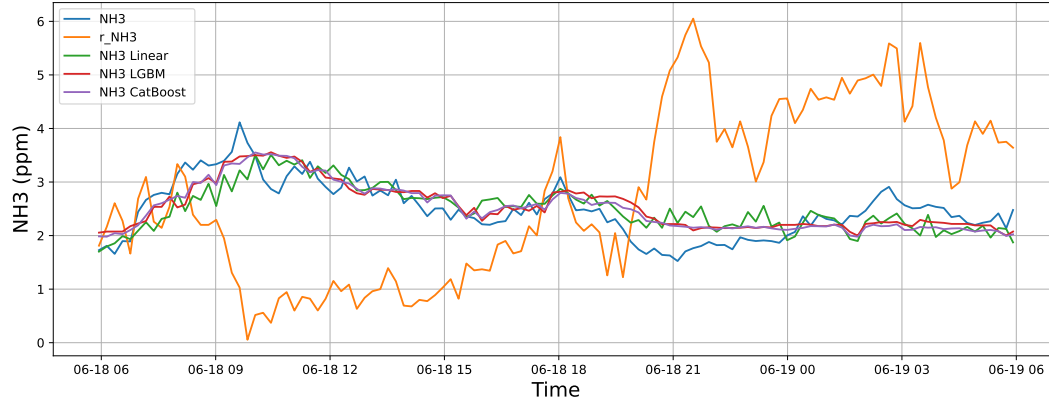


4.2 Compare plots

4.2.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

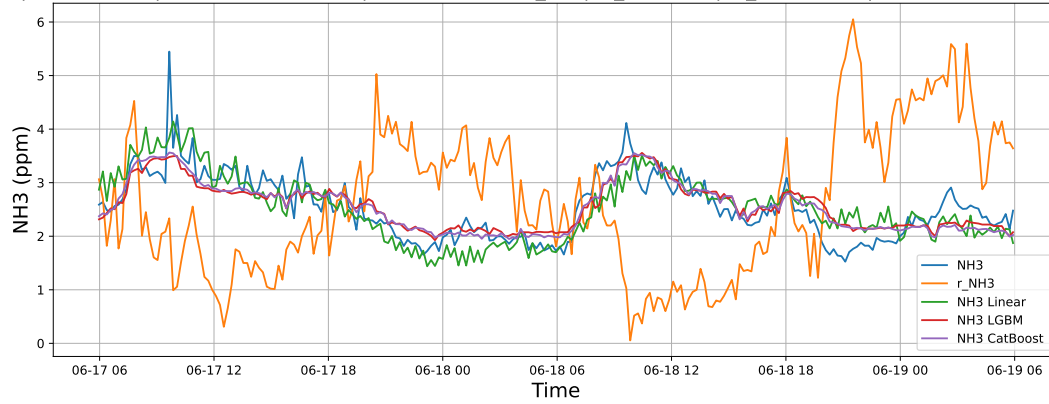
4.2.1.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



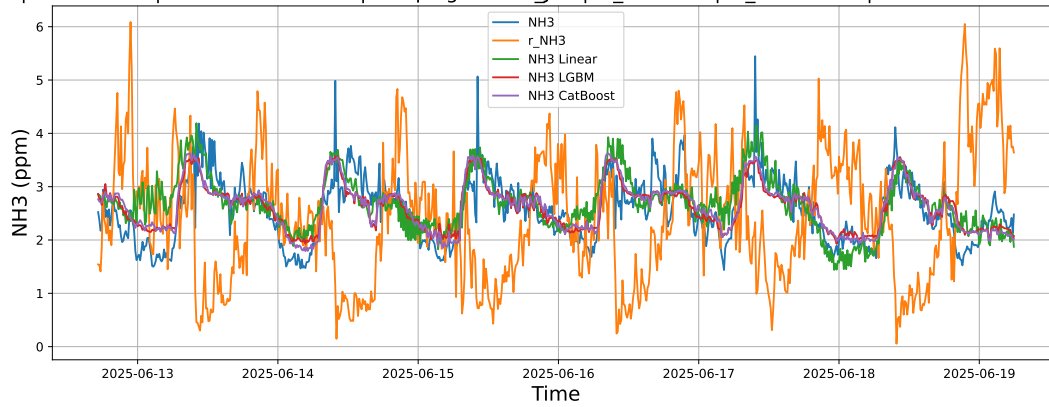
4.2.1.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



4.2.1.3 Time window : ALL

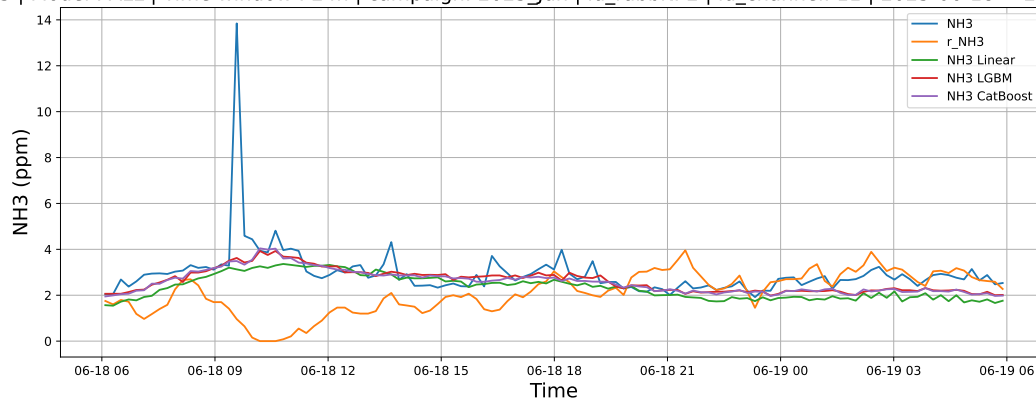
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



4.2.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

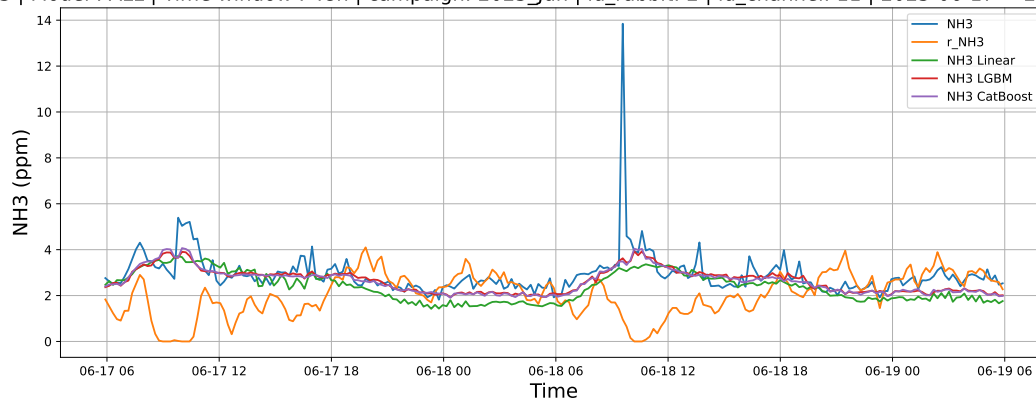
4.2.2.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



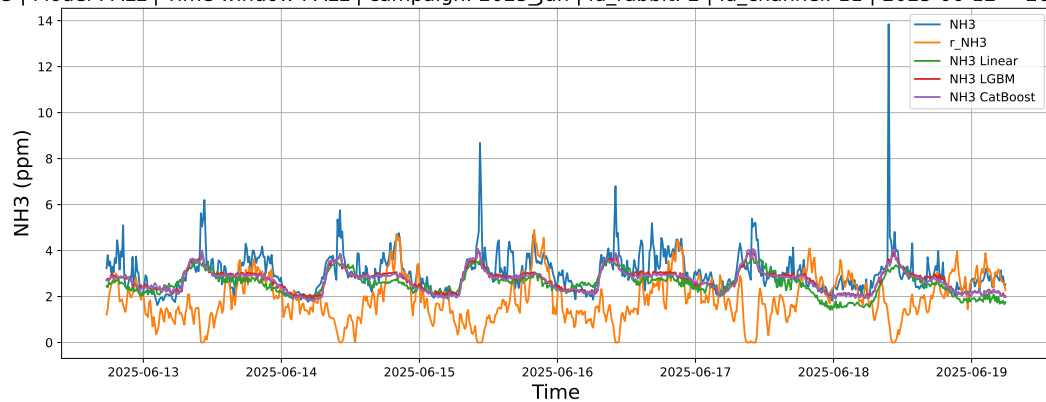
4.2.2.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



4.2.2.3 Time window : ALL

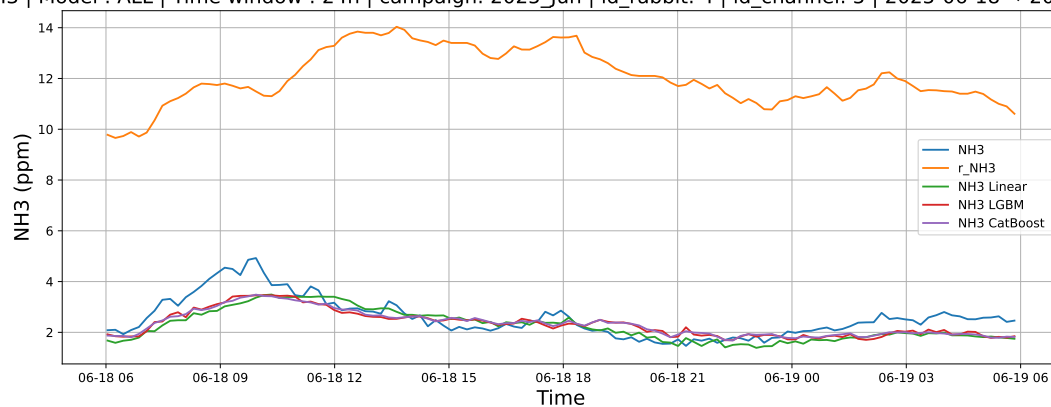
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



4.2.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

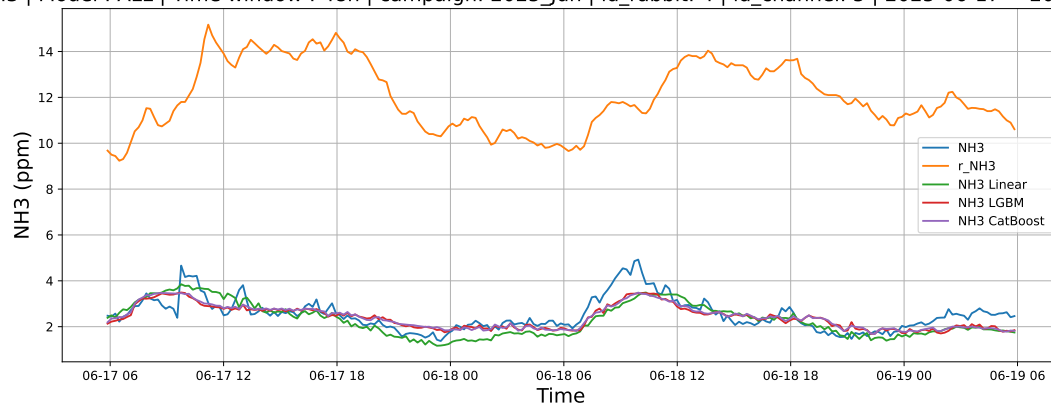
4.2.3.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



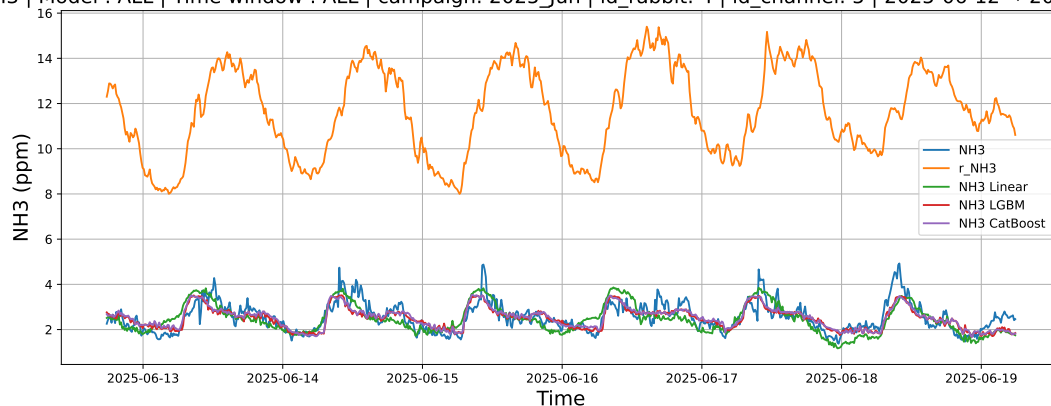
4.2.3.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



4.2.3.3 Time window : ALL

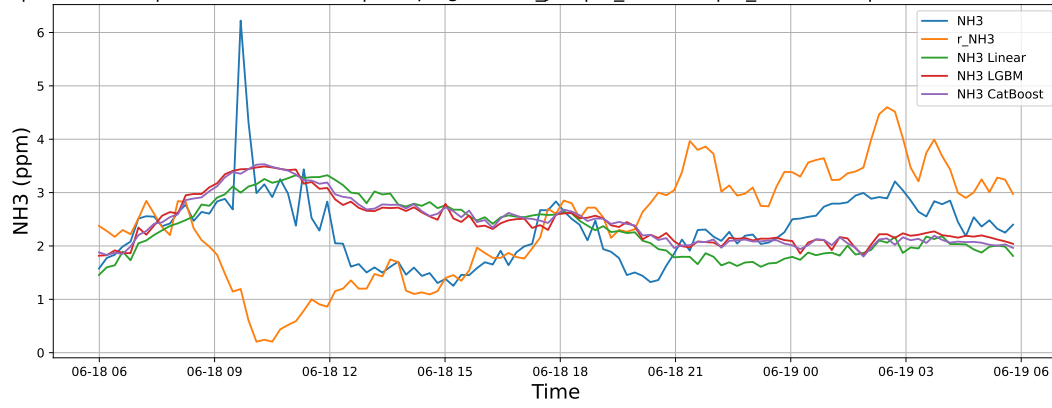
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



4.2.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

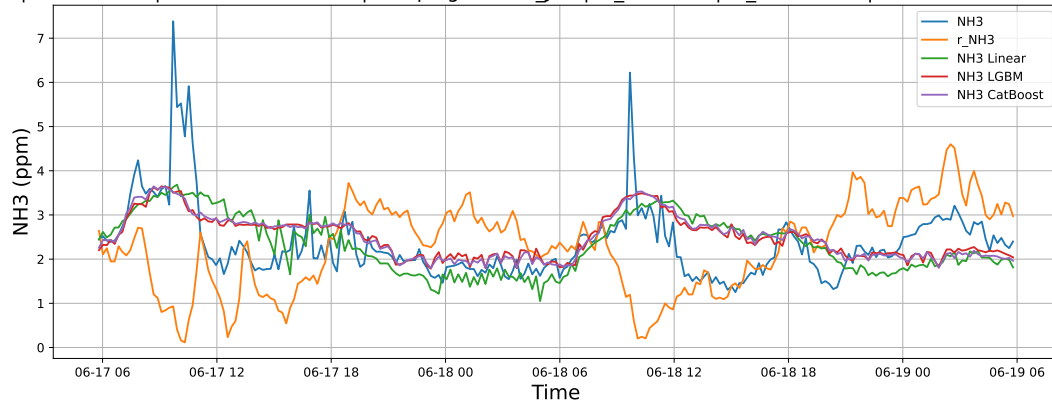
4.2.4.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



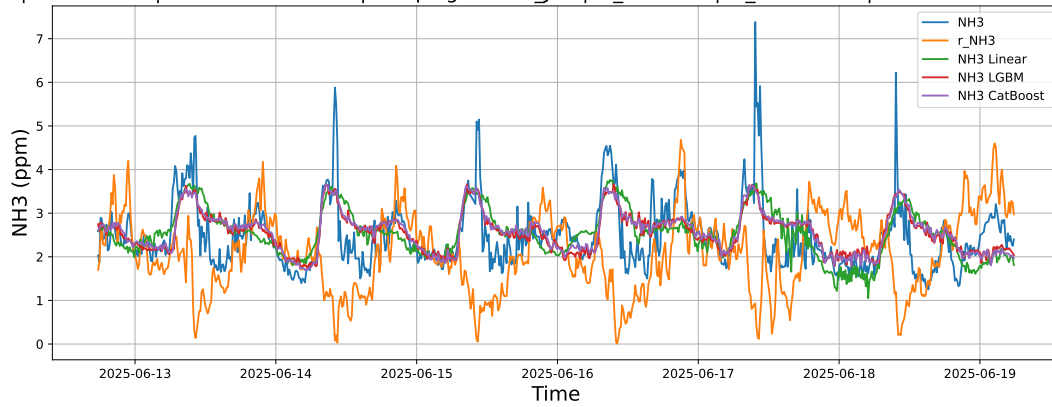
4.2.4.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



4.2.4.3 Time window : ALL

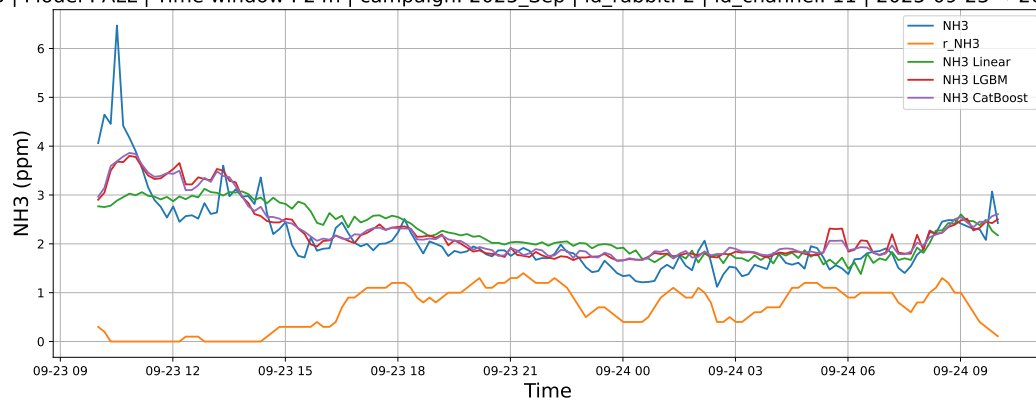
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



4.2.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

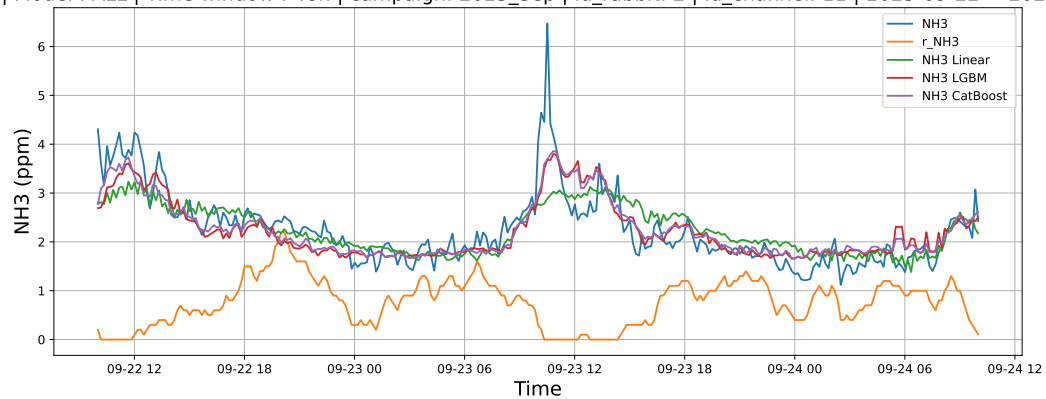
4.2.5.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



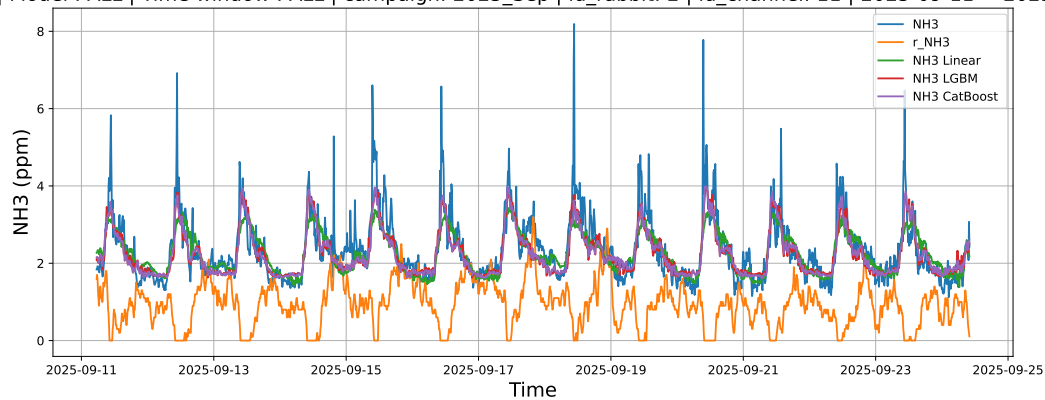
4.2.5.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



4.2.5.3 Time window : ALL

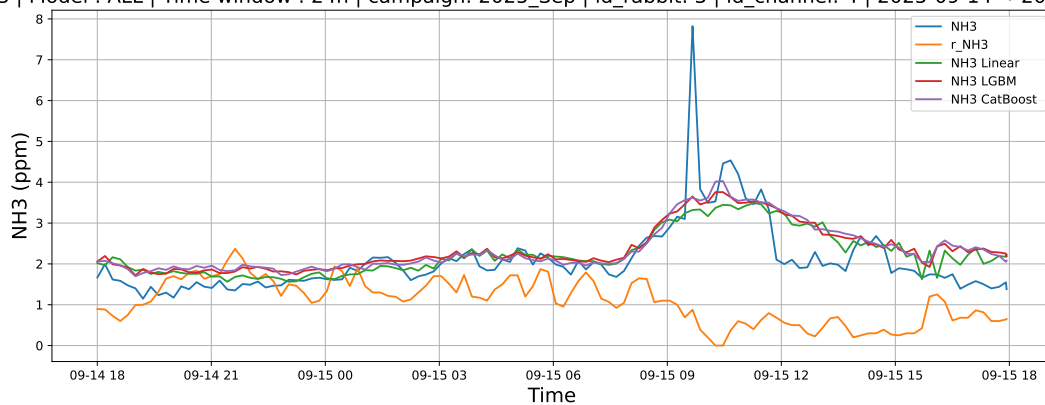
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



4.2.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

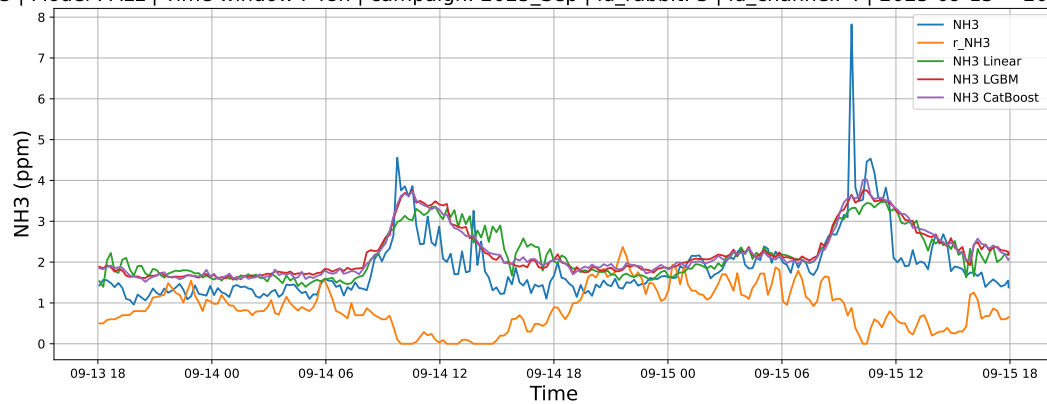
4.2.6.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



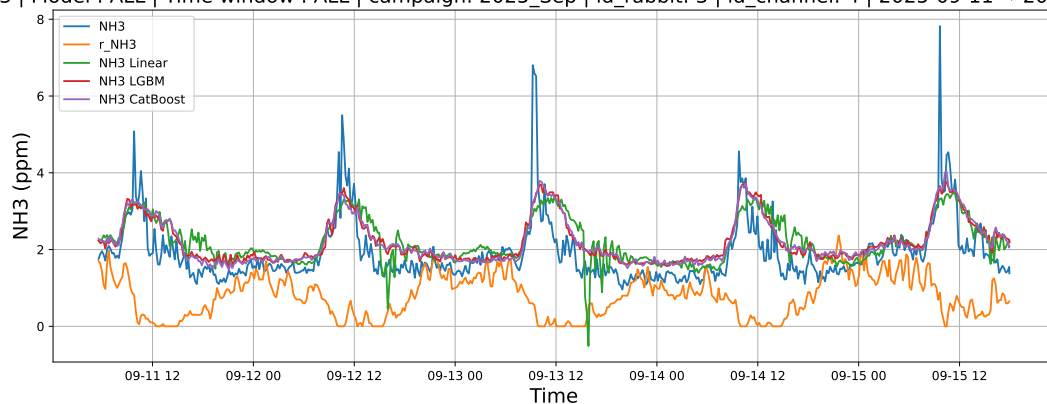
4.2.6.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



4.2.6.3 Time window : ALL

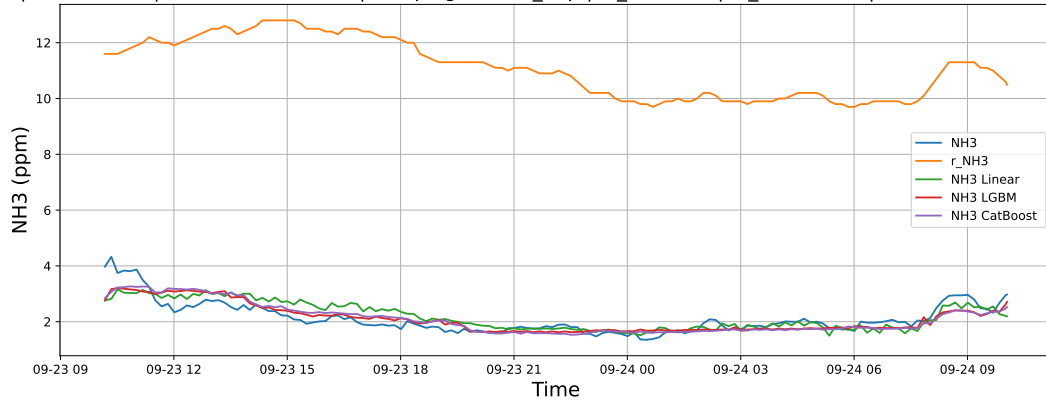
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



4.2.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

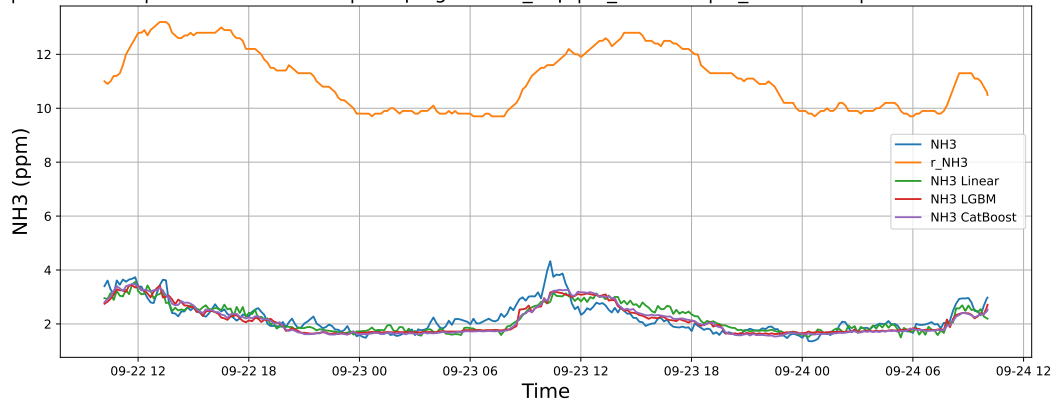
4.2.7.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



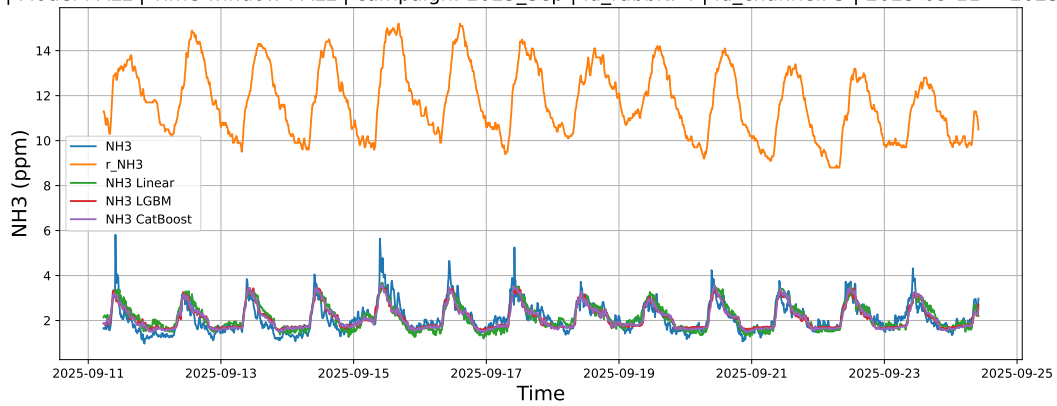
4.2.7.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



4.2.7.3 Time window : ALL

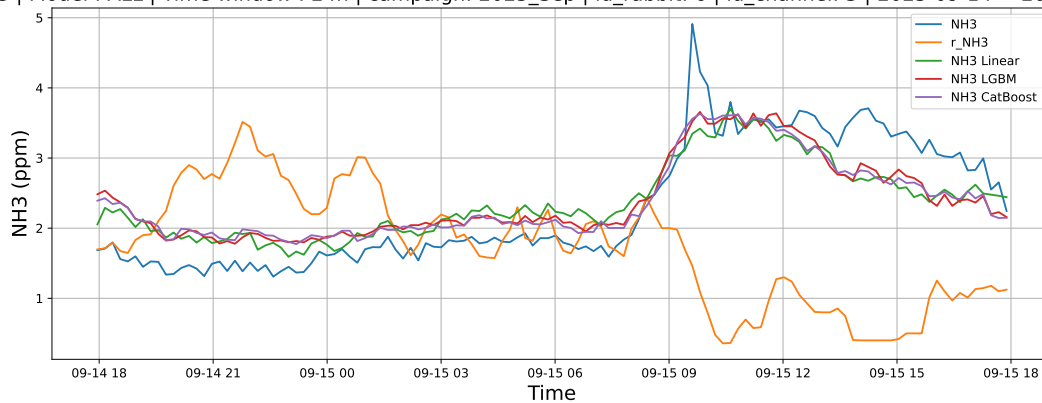
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



4.2.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

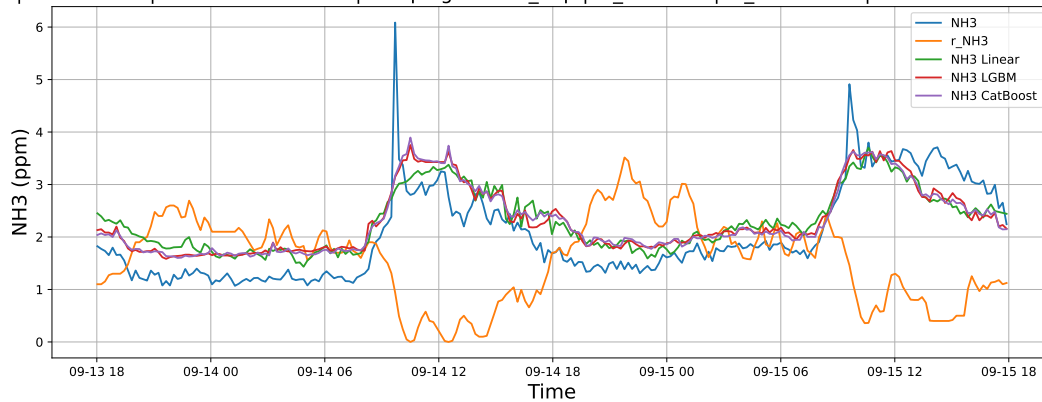
4.2.8.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



4.2.8.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



4.2.8.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

